COMPTE RENDU GÉNÉRAL

Administration Windows Server & Active Directory — M202

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Résumé des travaux

Ce document regroupe l'ensemble des travaux pratiques réalisés sur Windows Server / Active Directory, organisés en deux parties selon la méthode utilisée : interface graphique (GUI) et ligne de commande PowerShell (Server Core).

Partie A — Interface graphique (GUI)

- A.1 — Déploiement d'un contrôleur de domaine AD DS (NTIC Solutions)
- A.2 — Installation d'un contrôleur de domaine en lecture seule (RODC)
- A.3 — Administration des unités d'organisation (OU) — volet graphique

Partie B — Ligne de commande / PowerShell (Server Core)

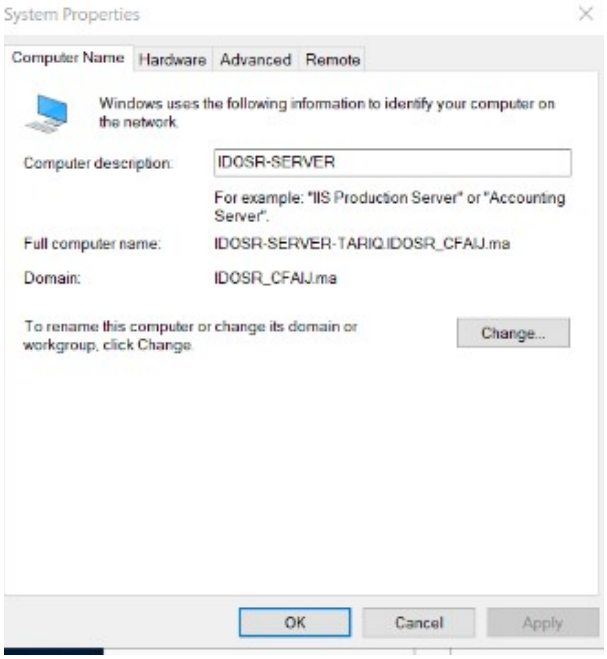
- B.1 — Mise en place d'un contrôleur de domaine avec PowerShell
- B.2 — TP N°9 : Gérer les objets Active Directory en PowerShell
- B.3 — Administration des unités d'organisation (OU) — volet PowerShell
- B.4 — Création de groupes et utilisateurs (TECHLOG-CORP)
- B.5 — Réinstallation d'un RODC via PowerShell (Server Core)

PARTIE A — INTERFACE GRAPHIQUE (GUI)

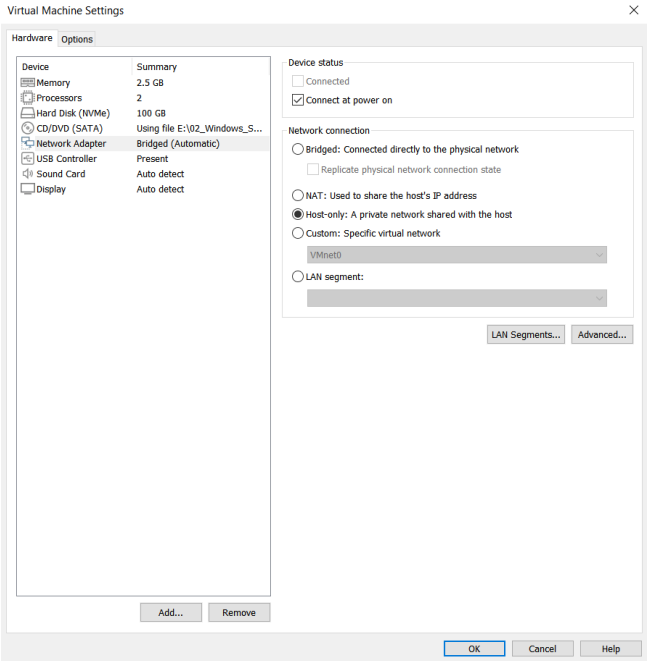
A.1 Déploiement d'un contrôleur de domaine AD DS

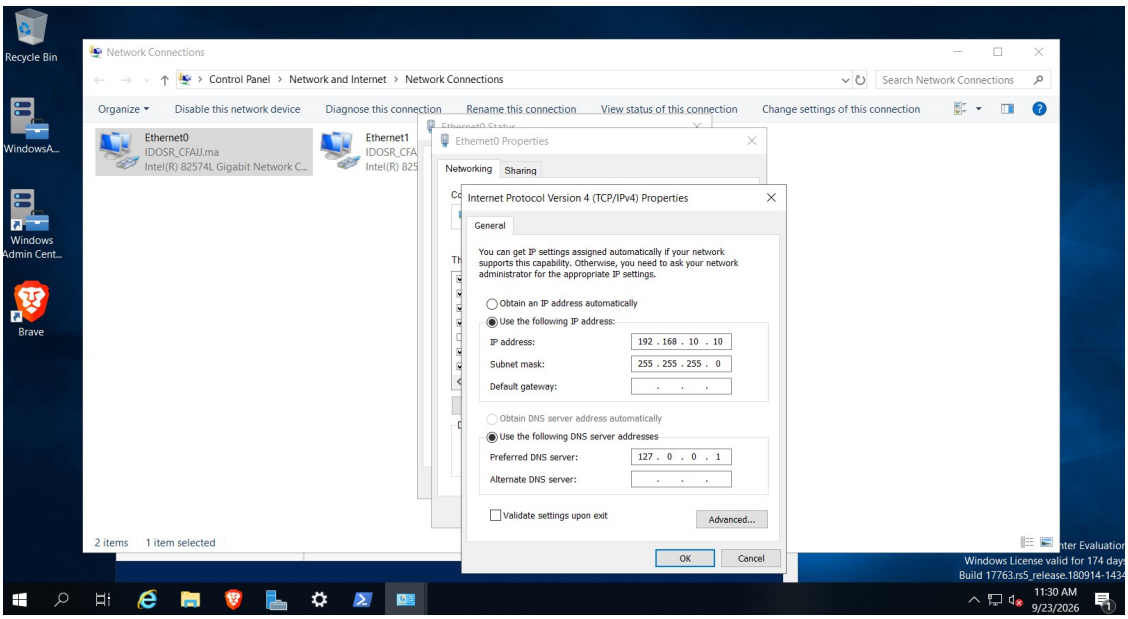
Problème 1 : Préparation et normalisation du serveur hôte

1. Modifiez le nom de l'ordinateur serveur pour qu'il devienne DC-NTIC.

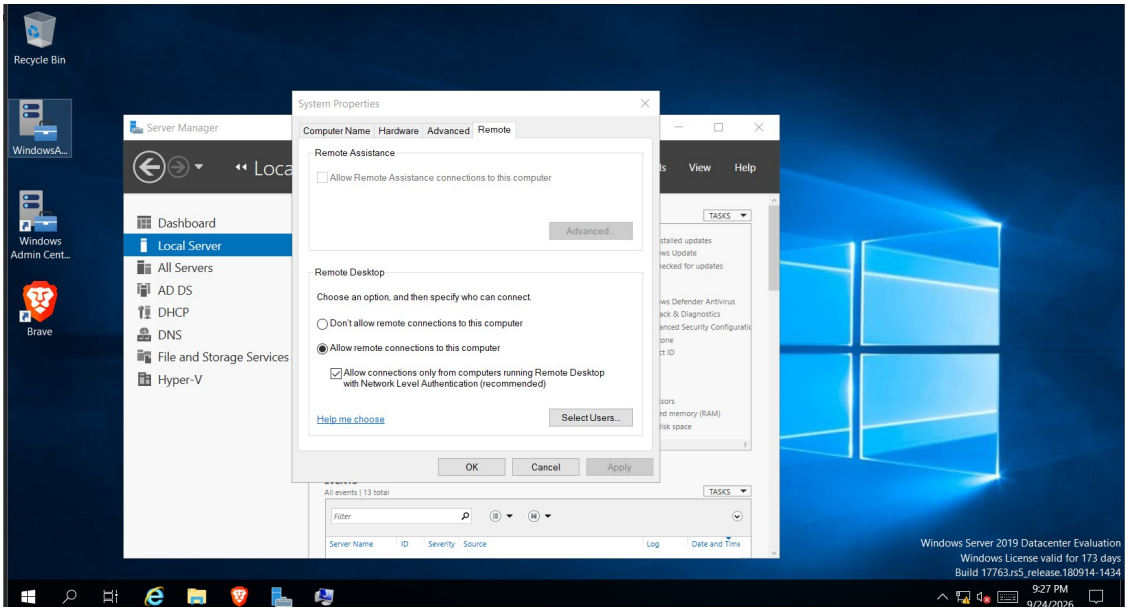


2. Configurez l'adresse IP statique 192.168.10.10/24 ainsi que le DNS préféré pointant vers lui-même.

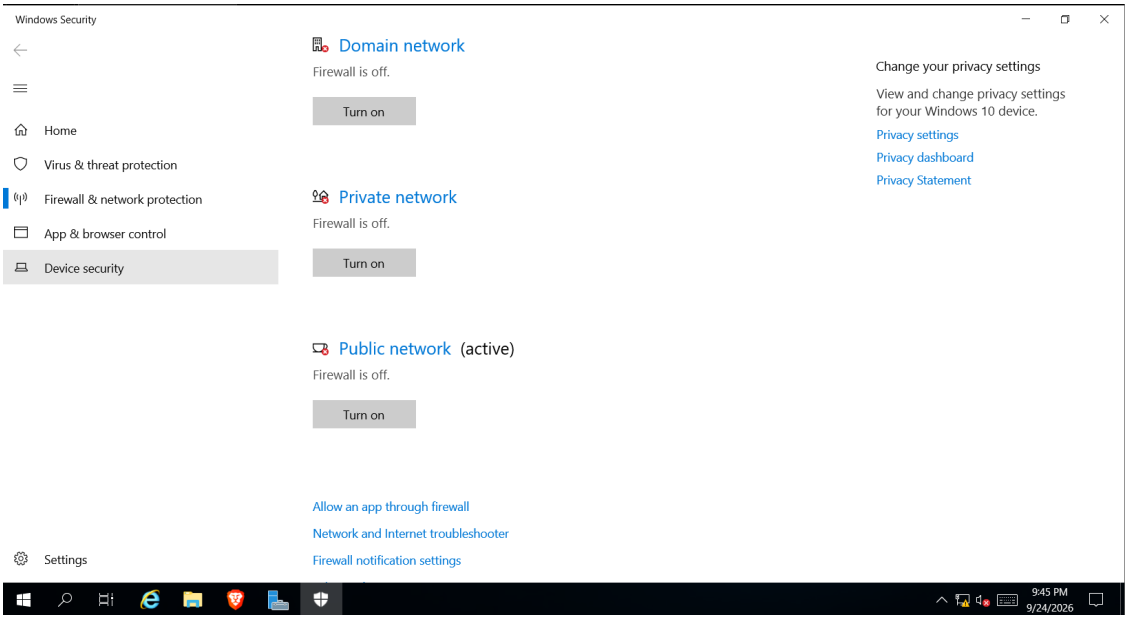




3. Activez le Bureau à distance (RDP).

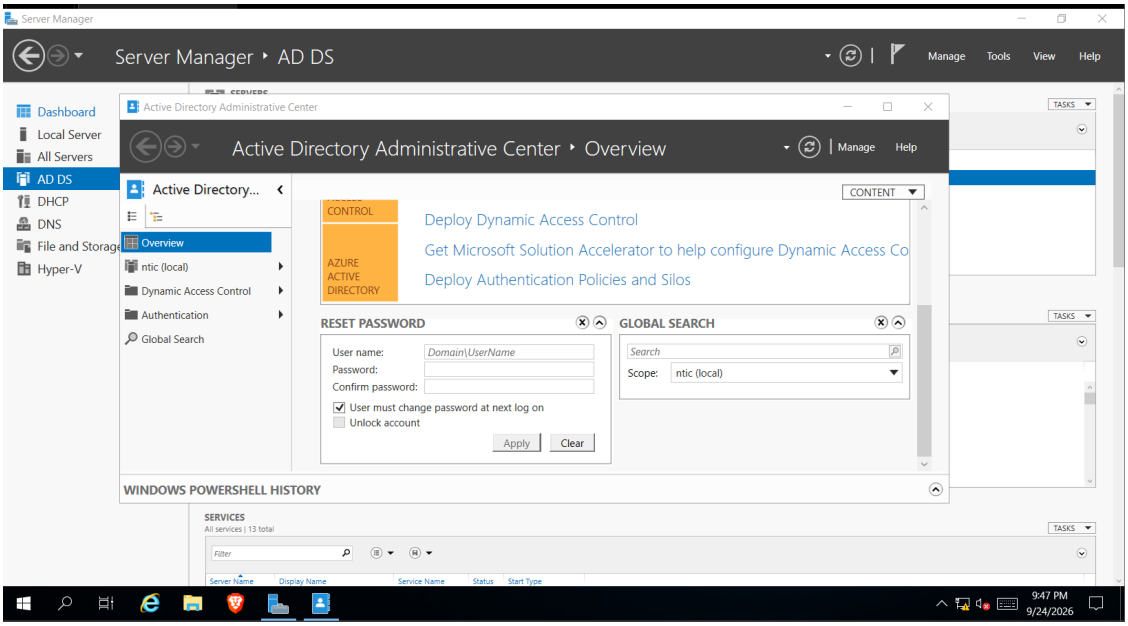


4. Configurez une règle dans le pare-feu Windows pour autoriser les requêtes ICMP Echo (Ping) entrantes.

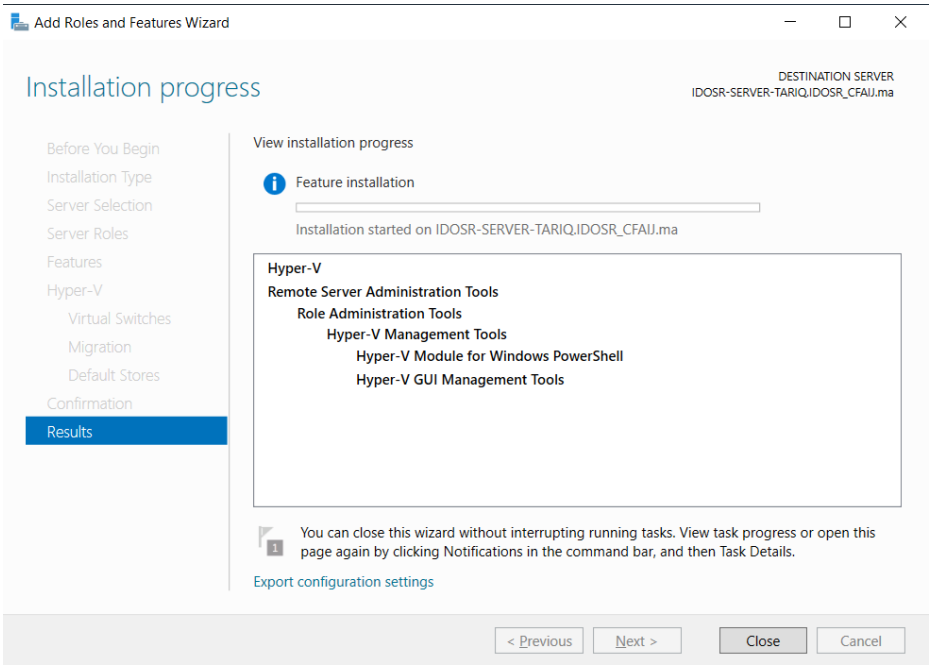


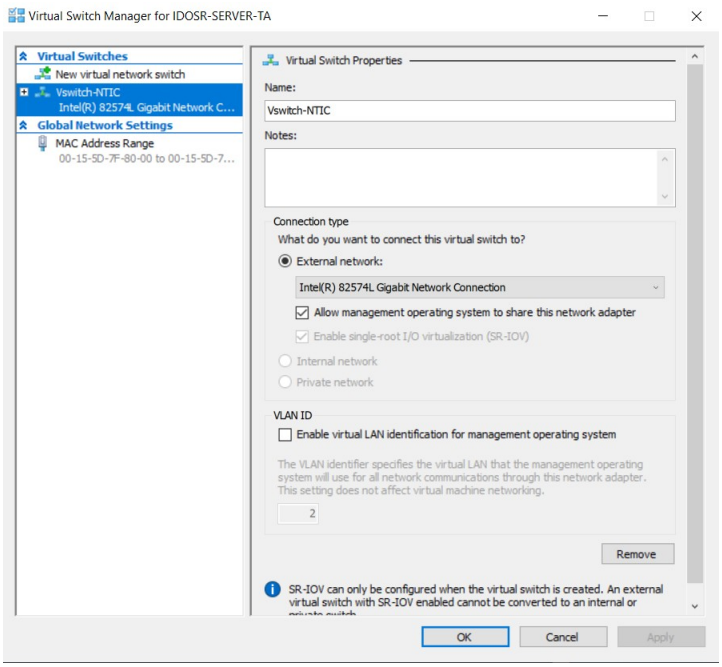
Problème 2 : Déploiement des services de domaine AD DS et du DNS

Installation du rôle AD DS + RSAT, promotion en nouvelle forêt ntic.lan, niveau fonctionnel Windows Server 2019, mot de passe DSRM.



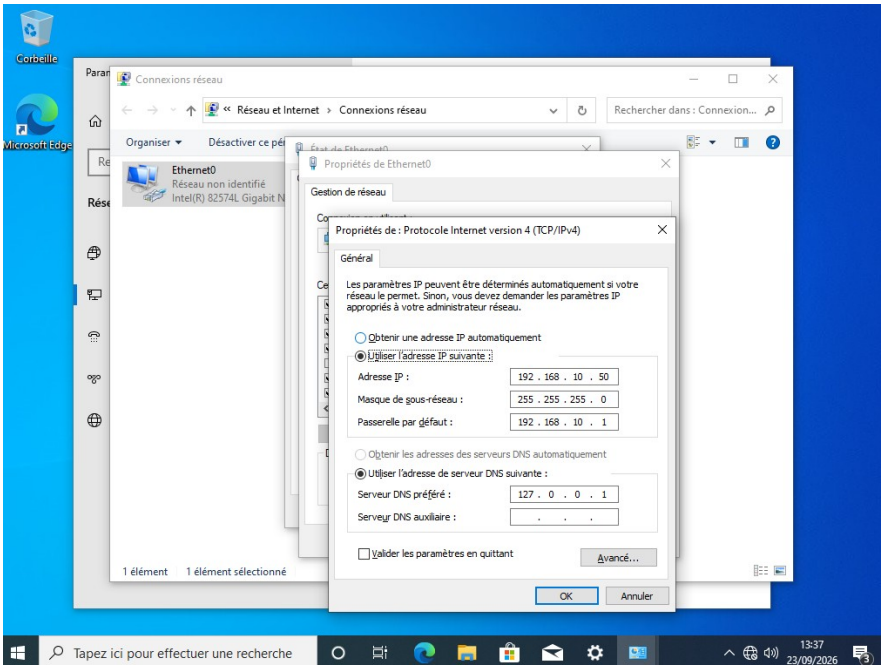
5. Créez un commutateur virtuel Hyper-V (VSwitch-NTIC).



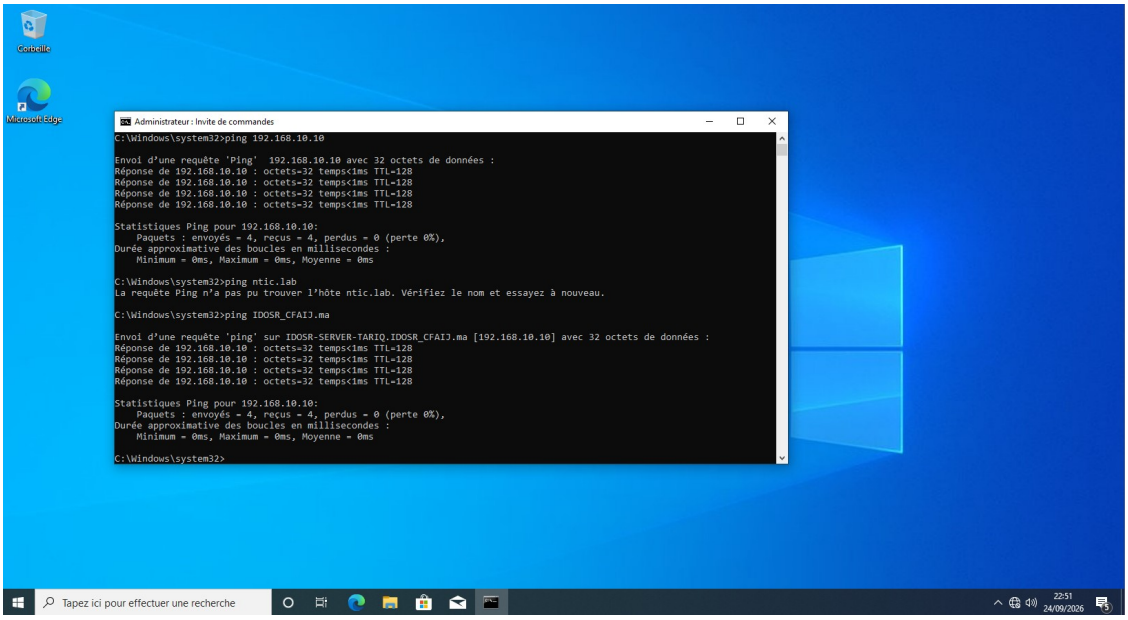


Problème 3 : Intégration du poste client Windows 10 dans le domaine

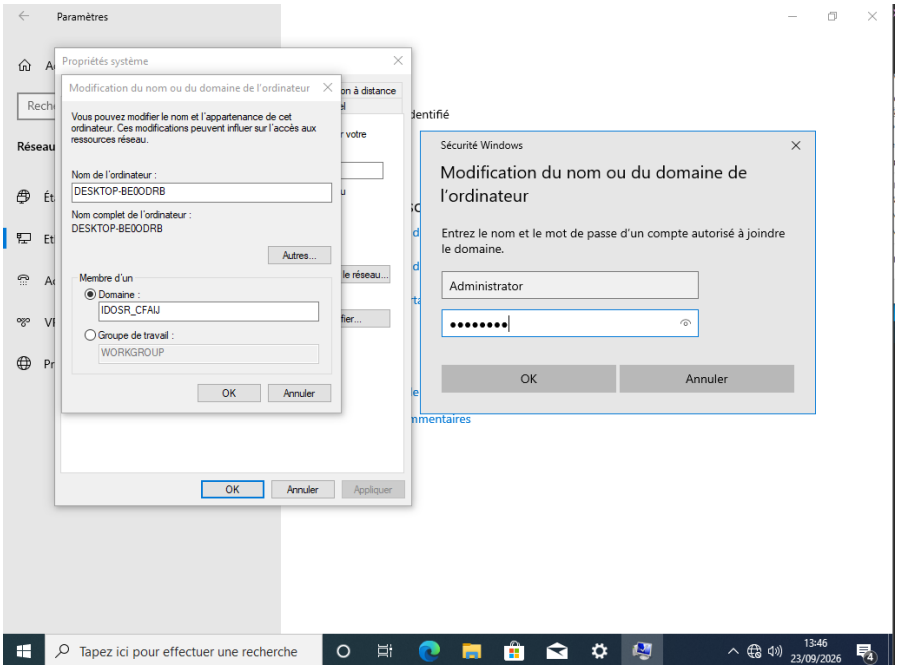
1. Configuration IP statique du poste (192.168.10.50, DNS 192.168.10.10).

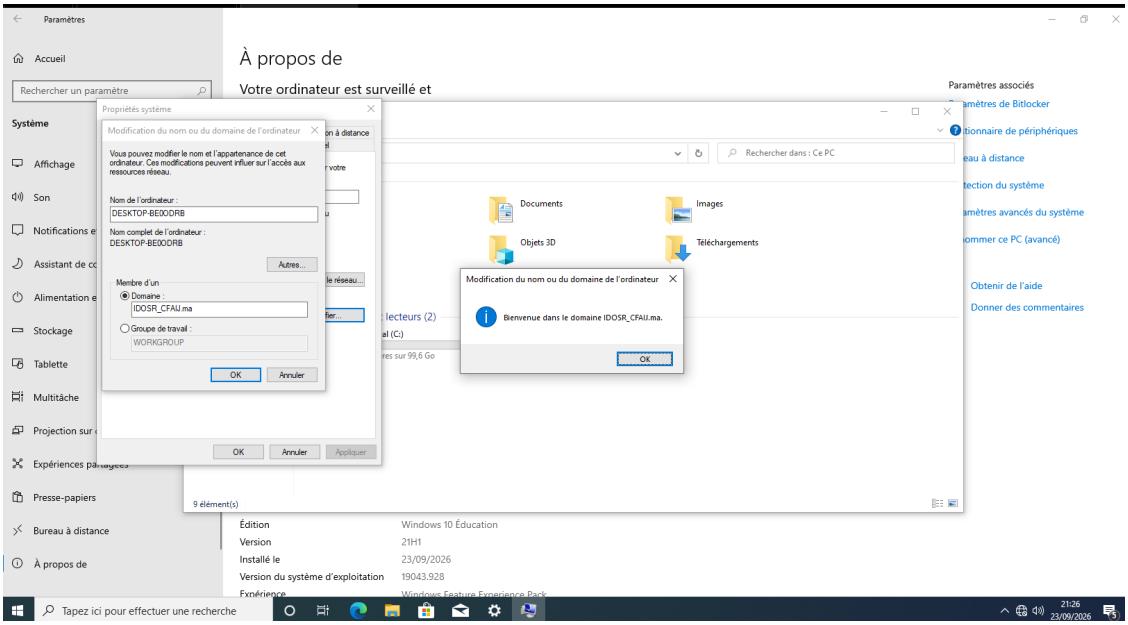


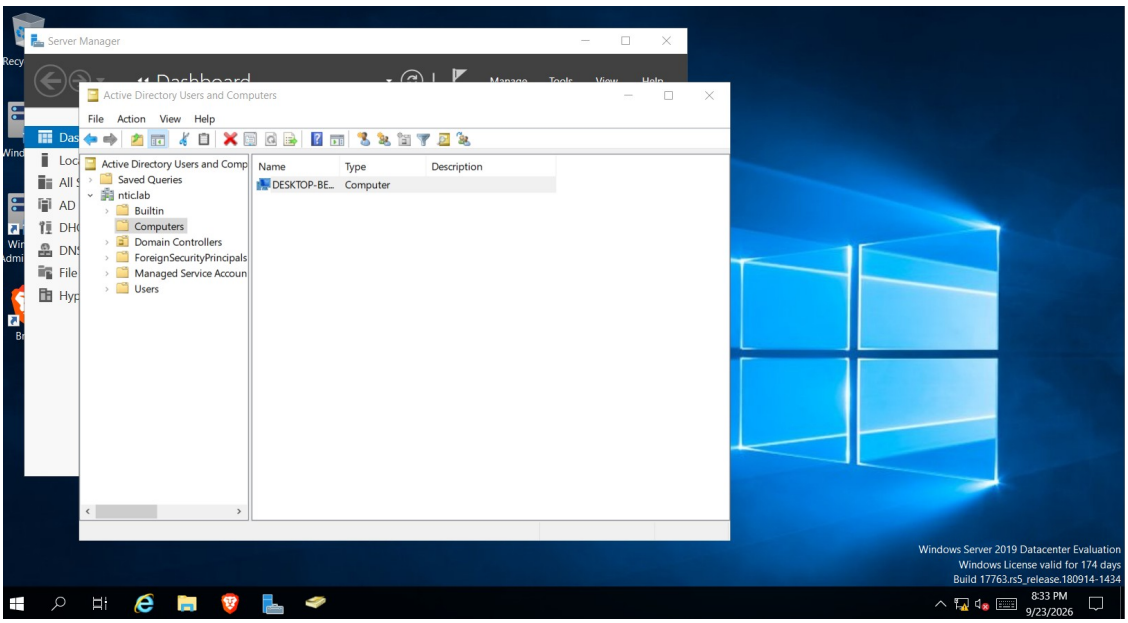
2. Test de résolution de nom et connectivité (ping ntic.lan).



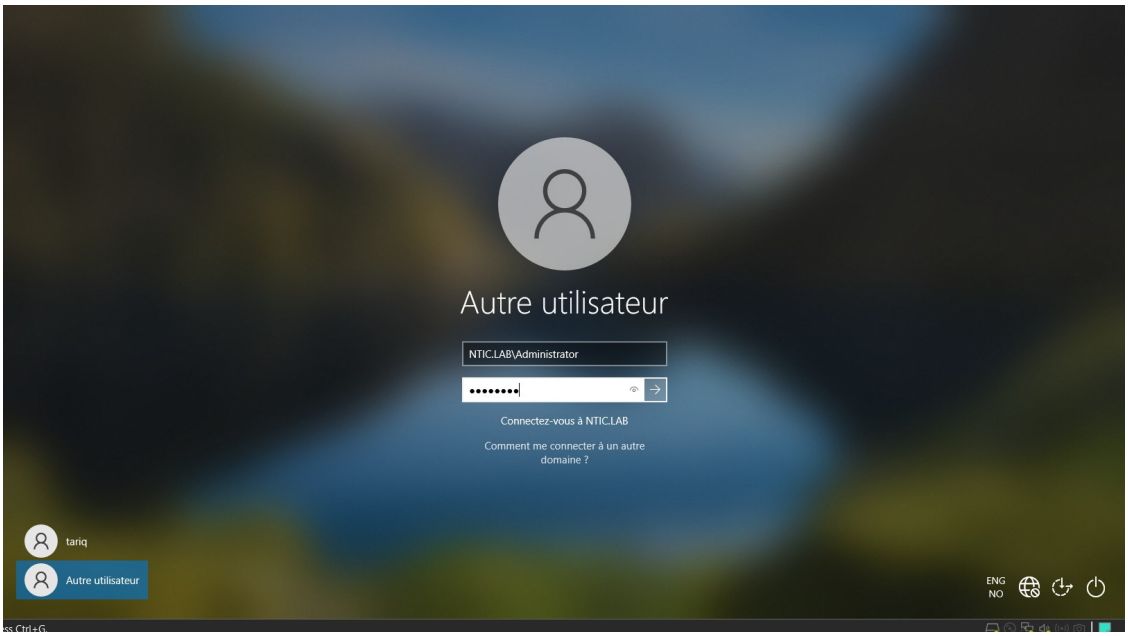
3. Jonction de CL-WIN10 au domaine ntic.lan.







4. Redémarrage et connexion avec NTIC\Administrateur.



Problème 4 : Surveillance avec System Insights

1. Installation de la fonctionnalité System Insights.

```
PS C:\Users\Administrator> Install-WindowsFeature -Name System-Insights -IncludeManagementTools

Success Restart Needed Exit Code      Feature Result
------- -------------- ---------      --------------
True    No             Success        {Feature Administration Tools, System Insi...
```

2. Liste des capacités de prévision.

```
PS C:\Users\Administrator> Get-InsightsCapability

Name                                 Description                                                  State   Status Last Run
----                                 -----------                                                  -----   ------ --------
CPU capacity forecasting             Forecasts CPU usage.                                         Enabled None
Networking capacity forecasting      Forecasts network usage for each network adapter.            Enabled None
Total storage consumption forecasting Forecasts total storage consumption across all local drives. Enabled None
Volume consumption forecasting       Forecasts storage consumption for each volume.               Enabled None
```

3. Prédiction manuelle sur la capacité de stockage.

```
PS C:\Users\Administrator> Invoke-InsightsCapability -Name "Total storage consumption forecasting"

Invoking a capability:
Invoking capability 'Total storage consumption forecasting' is a computationally expensive operation. Consider
scheduling periodic predictions during CPU downtime. The most recent prediction will be automatically stored on
completion. You can view the most recent prediction and associated data using the Get-InsightsCapabilityResult cmdlet.
Do you want to continue?
[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "Y"): Y
PS C:\Users\Administrator>
```

4. Résultats de l'analyse.

```
PS C:\Users\Administrator> Get-InsightsCapabilityResult -Name "Total storage consumption forecasting"

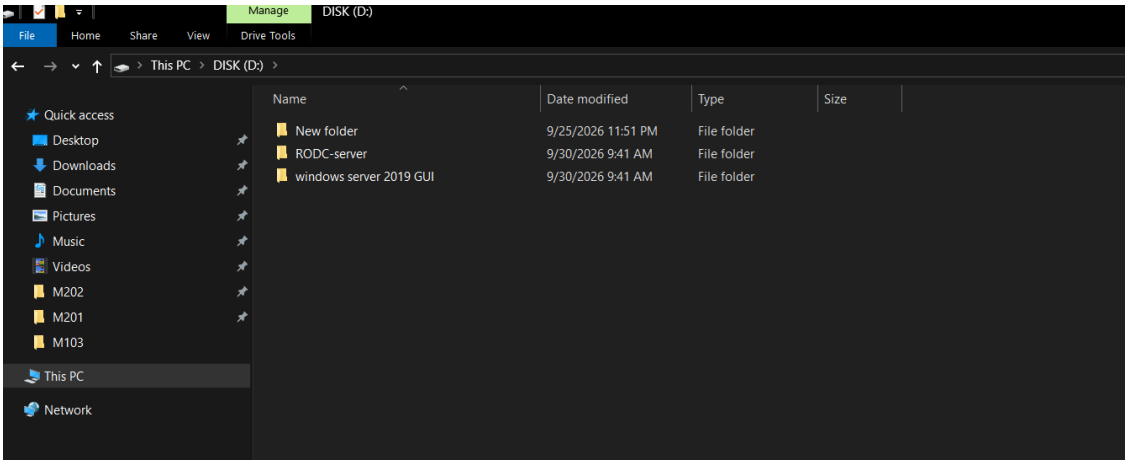
Name                                Status Status Description
----                                ------ ------------------
Total storage consumption forecasting None   Check back again later -  we don’t have enough recent data to make pred...
a


PS C:\Users\Administrator>
```

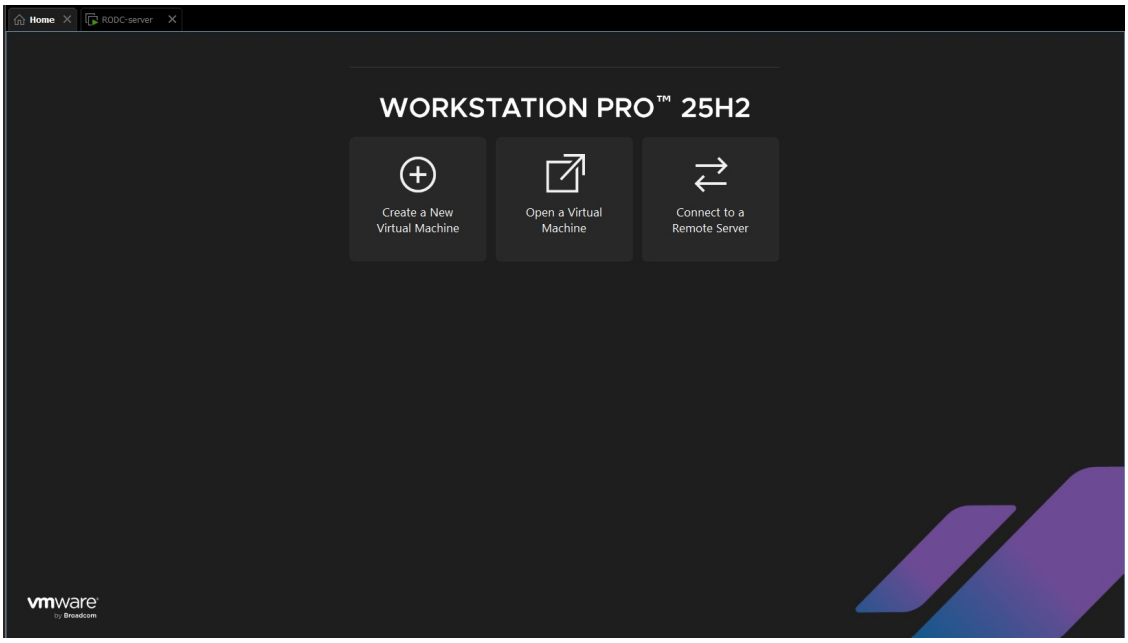
A.2 Installation d'un contrôleur de domaine en lecture seule (RODC)

1. Préparation de l'environnement (clonage de la VM)

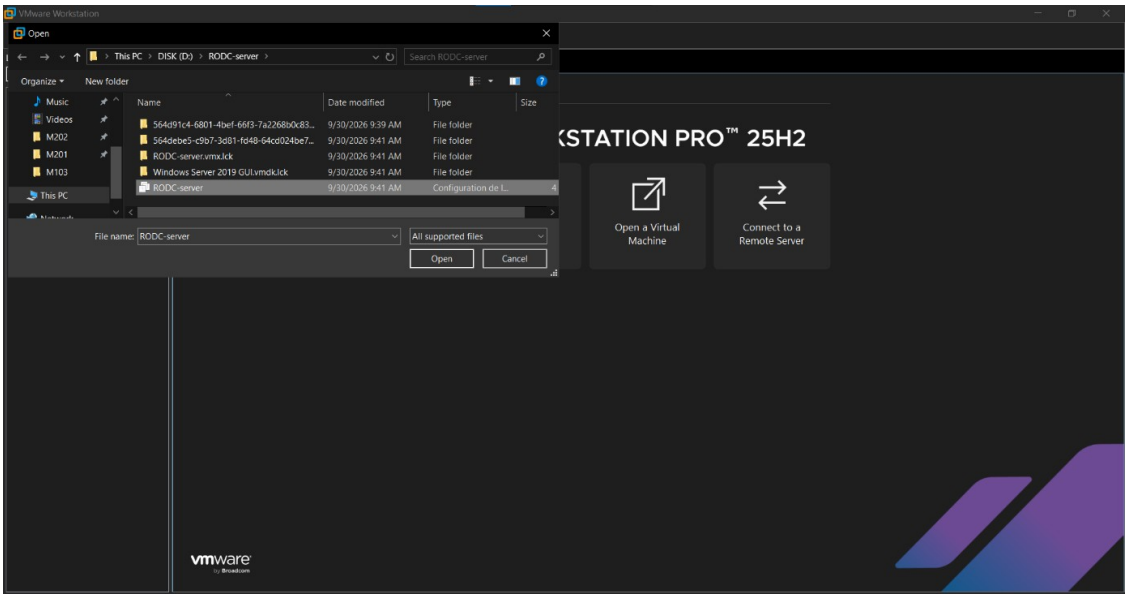
Le disque D: de l'hôte contient l'ISO de Windows Server 2019 ainsi que le dossier de la machine virtuelle clonée, destinée à devenir le futur RODC.



L'hyperviseur VMware Workstation Pro est démarré pour accueillir la machine virtuelle clonée.



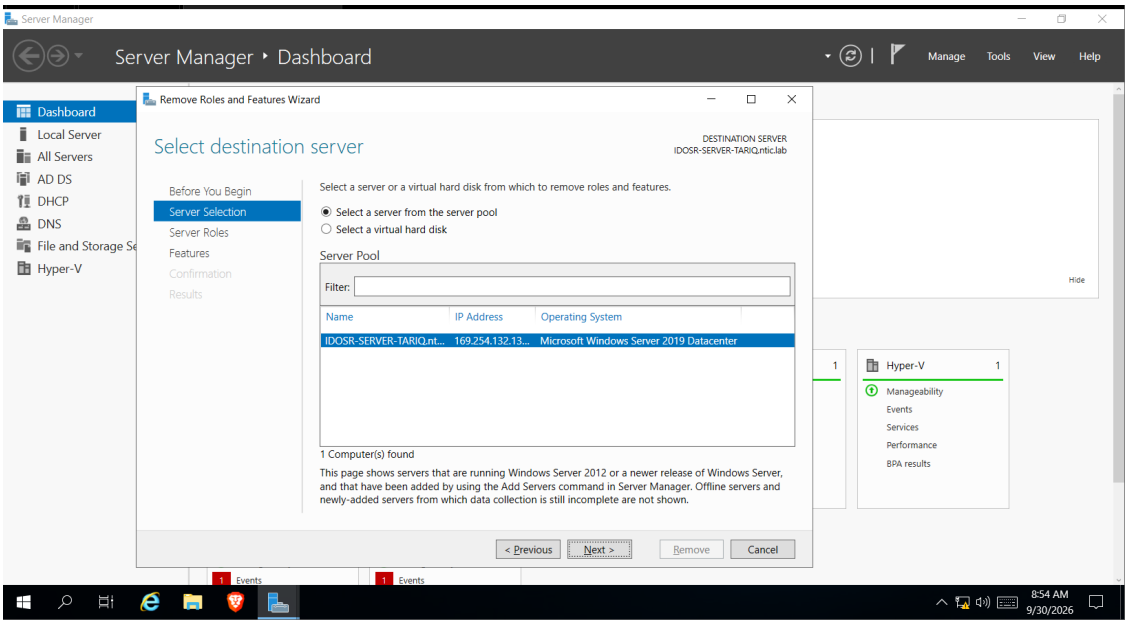
La machine virtuelle clonée est ouverte depuis le dossier D:\RODC-server, puis démarrée (écran de verrouillage).

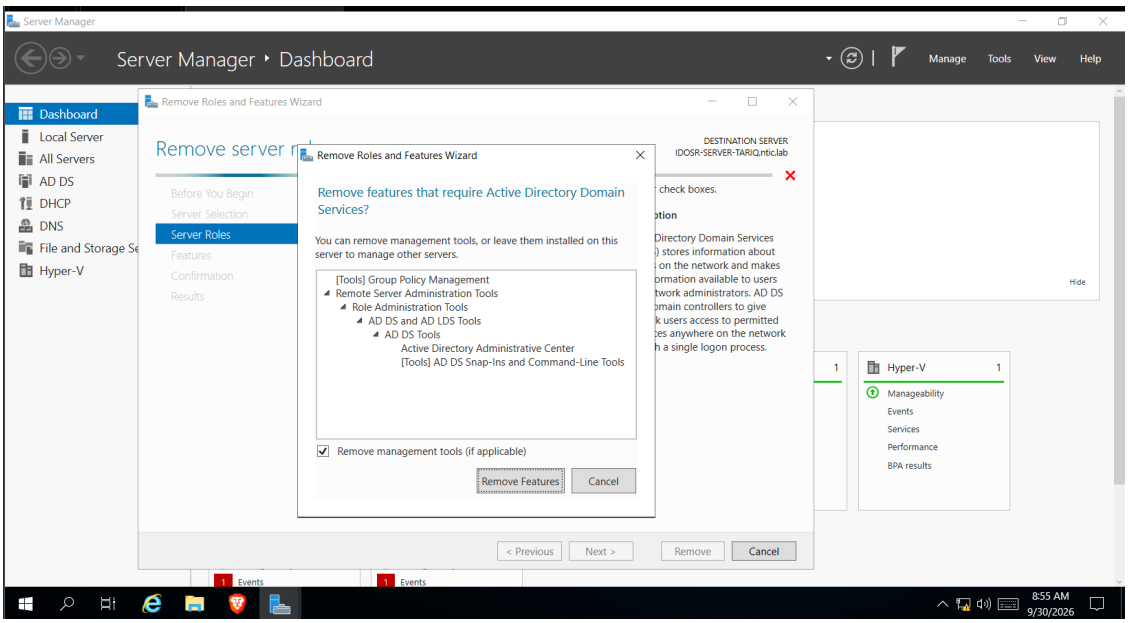




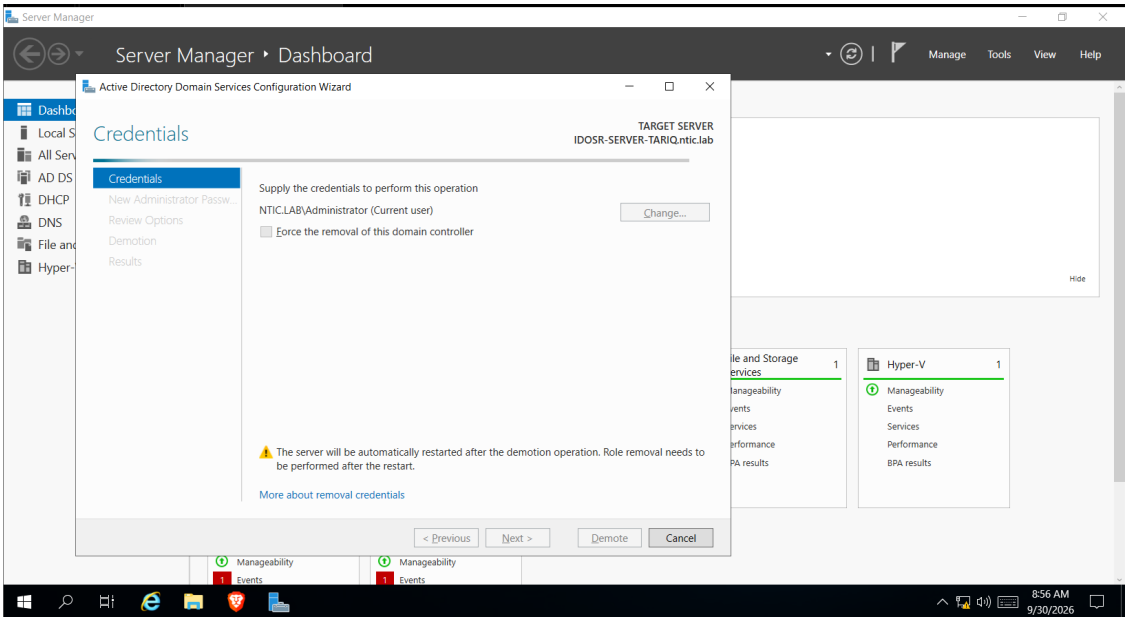
2. Suppression d'AD DS sur le serveur cloné (rétrogradation)

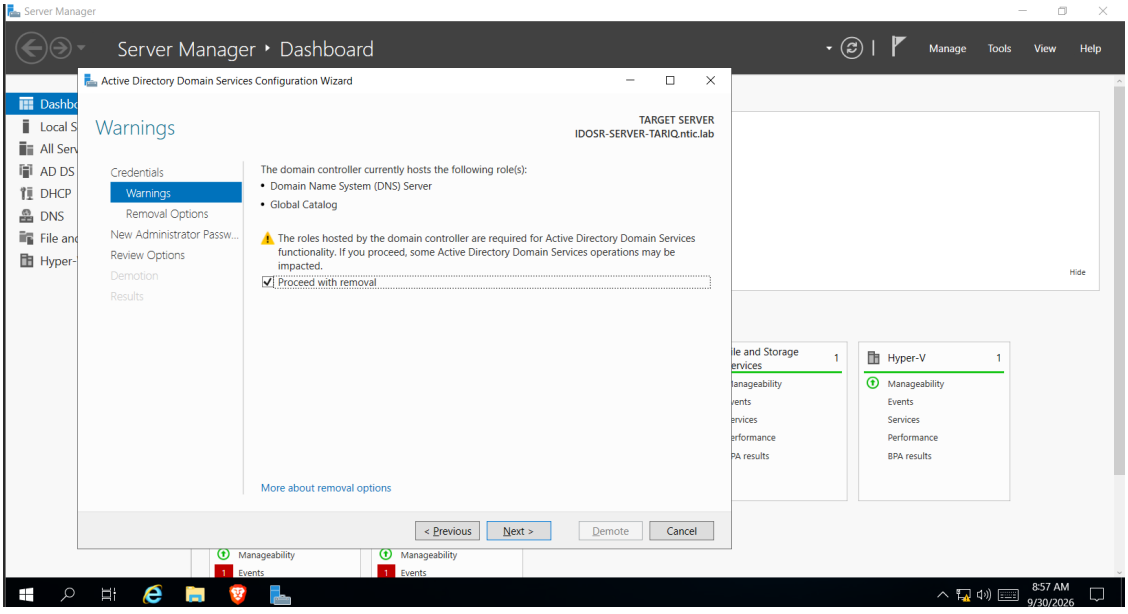
Le clone possédant déjà le rôle AD DS du serveur d'origine, l'assistant « Remove Roles and Features » retire ce rôle avant de pouvoir reconfigurer le serveur en RODC.

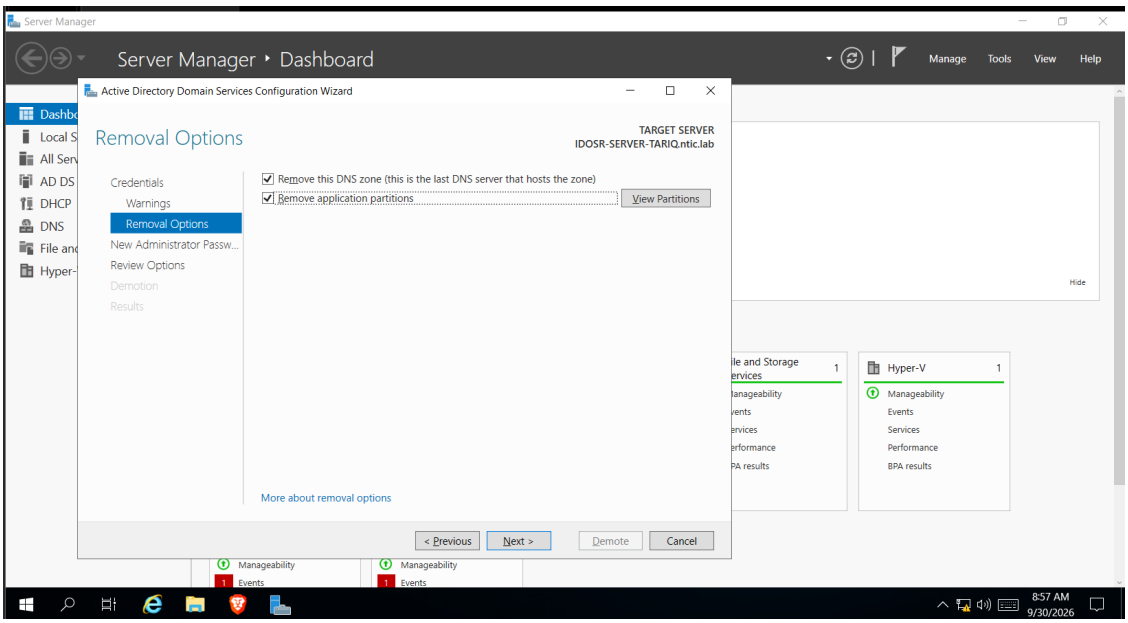




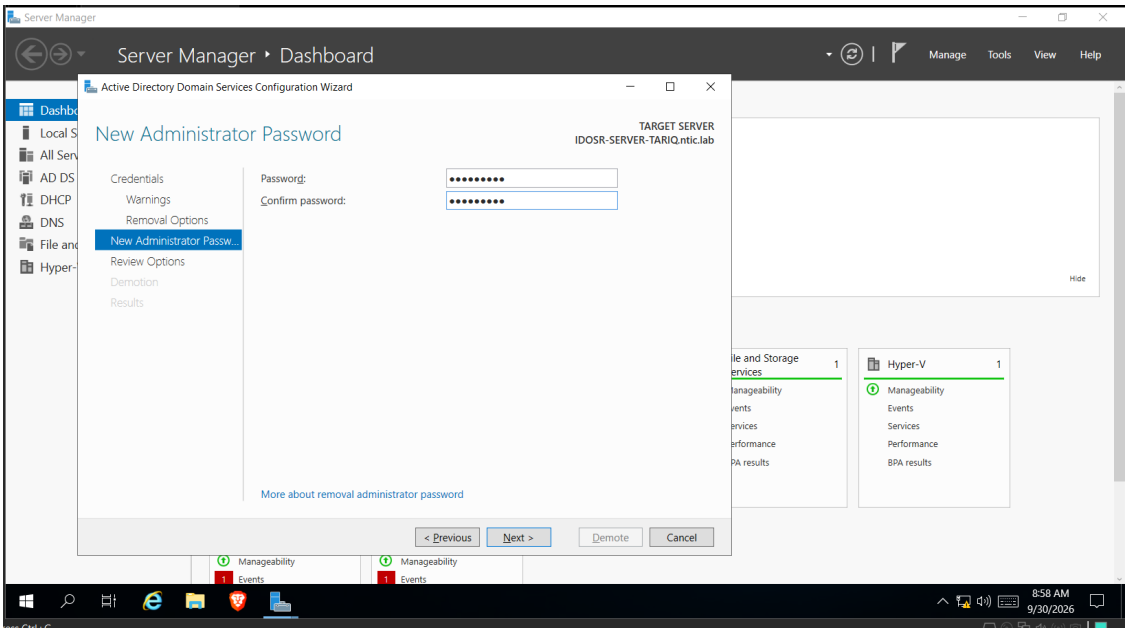
L'assistant de rétrogradation demande les identifiants, affiche les avertissements puis les options de suppression.







Un nouveau mot de passe administrateur local est défini, ce qui finalise la rétrogradation.

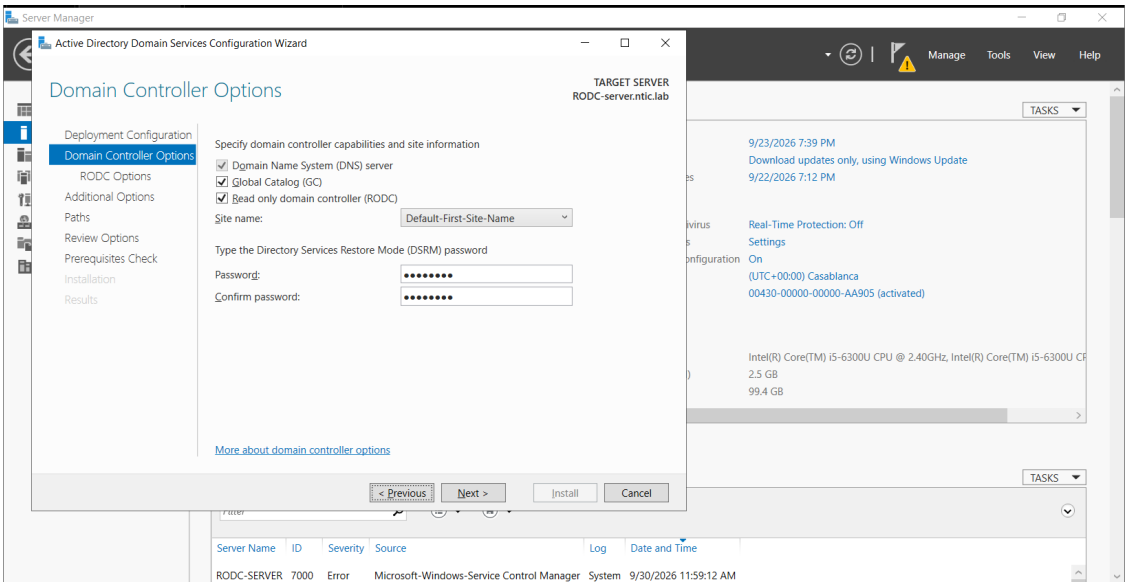


3. Renommage du serveur et changement d'adresse IP

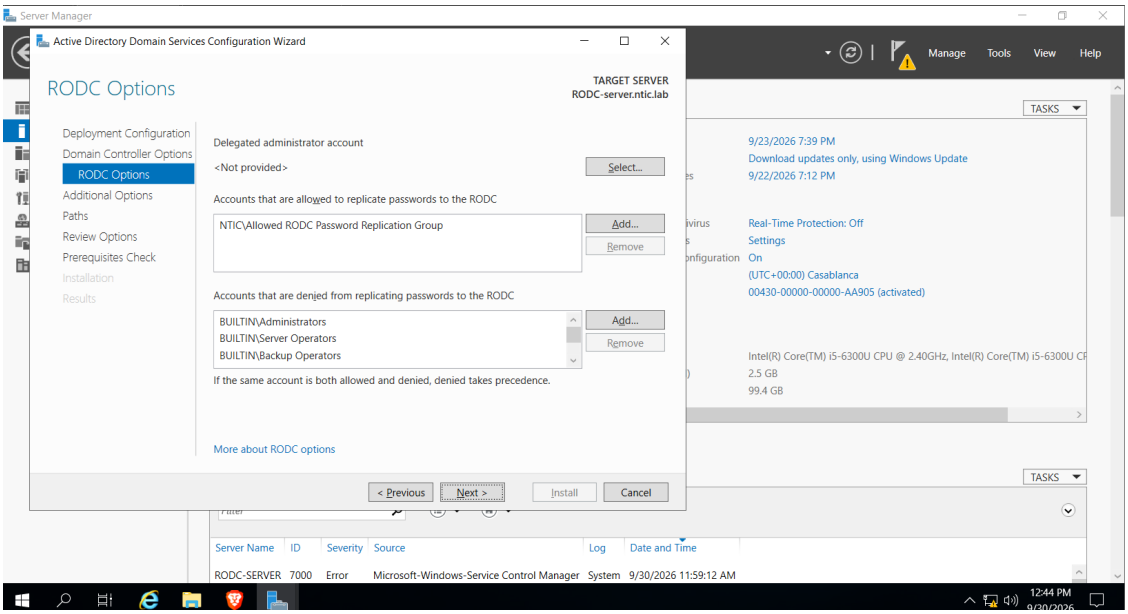
Le serveur cloné est renommé RODC-SERVER et une nouvelle adresse IP statique (192.168.10.11) lui est attribuée pour éviter tout conflit avec le contrôleur de domaine d'origine (voir Partie B.5 pour les commandes PowerShell équivalentes).

4. Installation et promotion en RODC

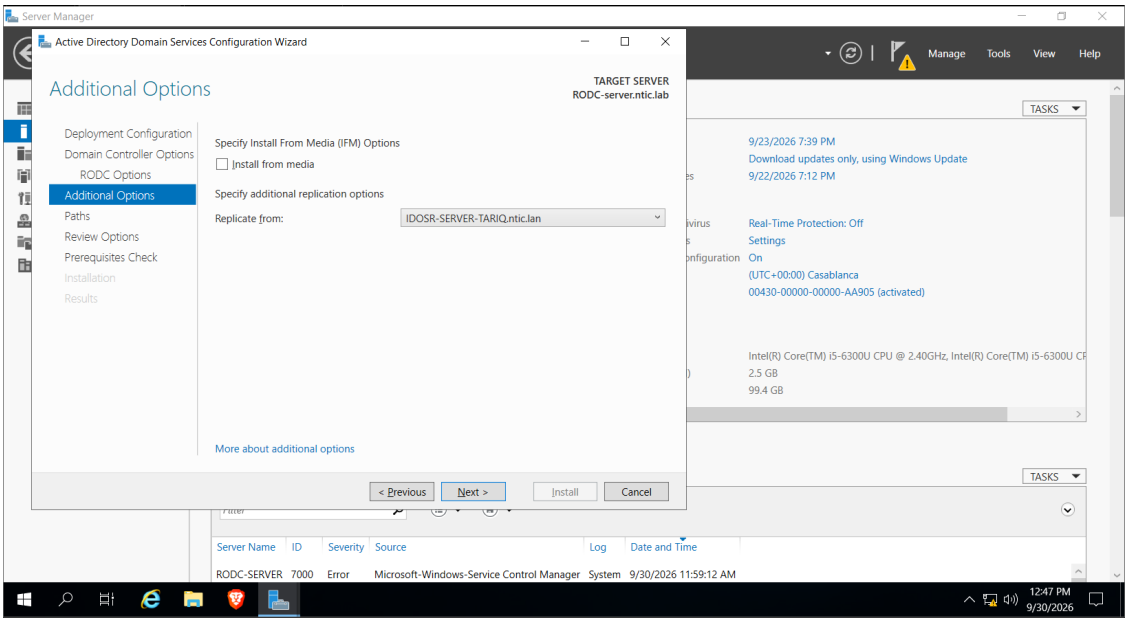
Dans l'assistant de configuration AD DS : DNS Server, Global Catalog et Read only domain controller (RODC) sont cochés, mot de passe DSRM défini.

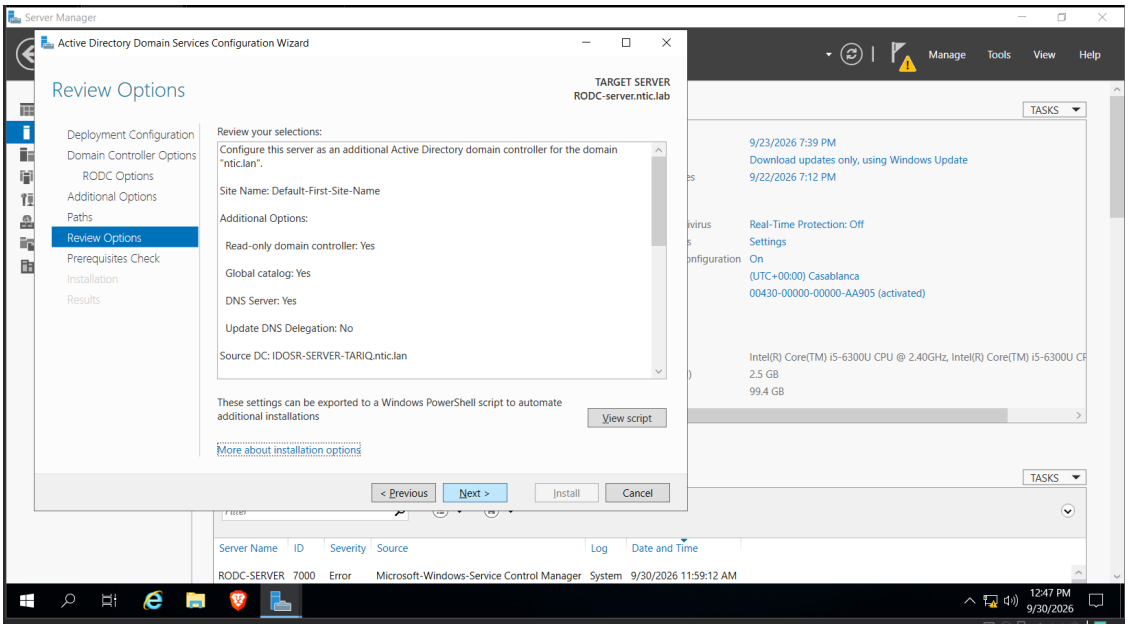


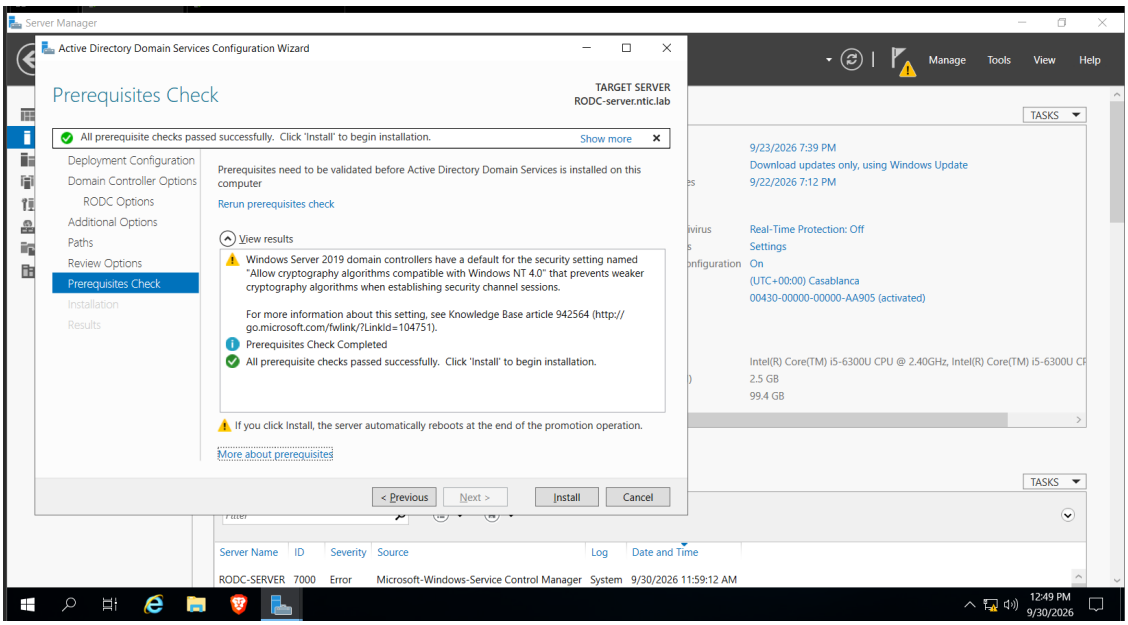
Compte administrateur délégué et groupes autorisés/refusés pour la réplication des mots de passe.

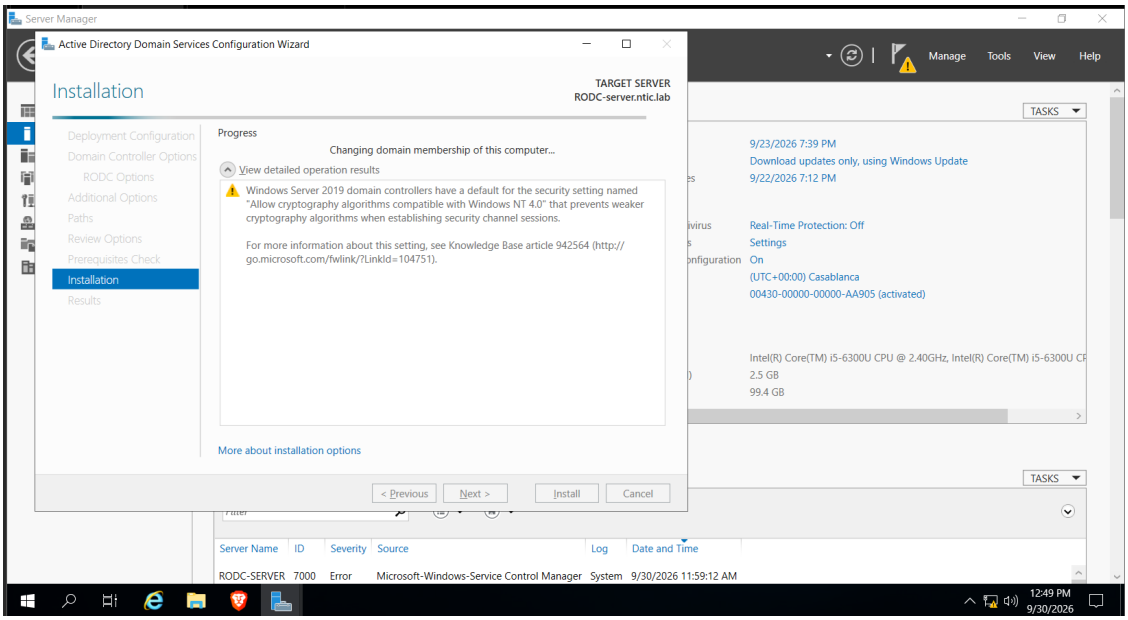


Contrôleur source de réplication, récapitulatif, vérification des prérequis, installation.

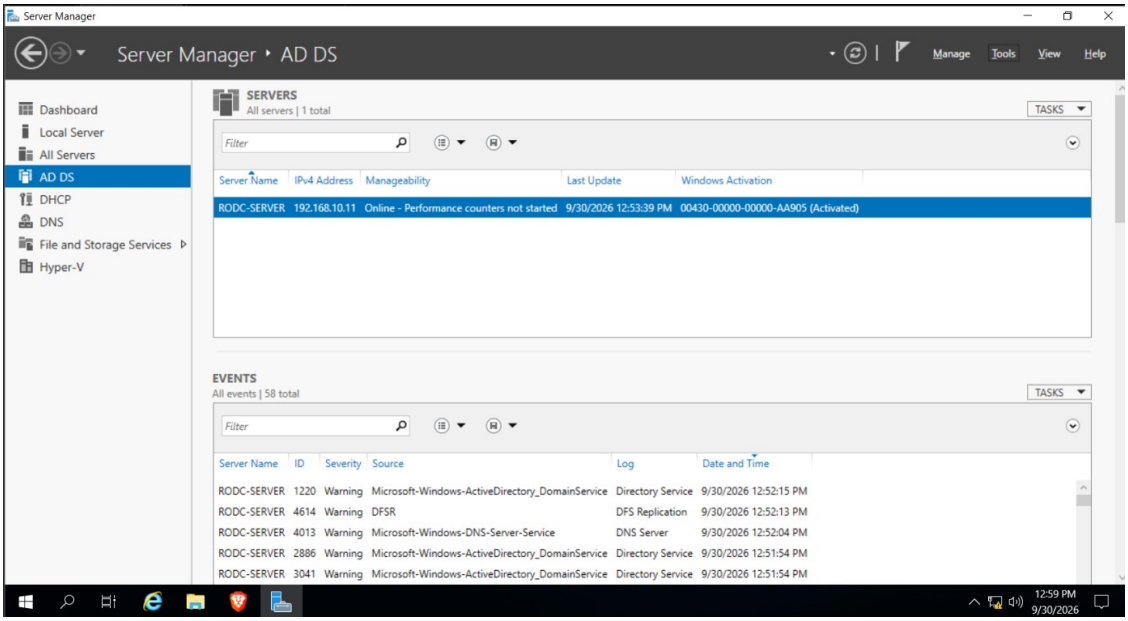








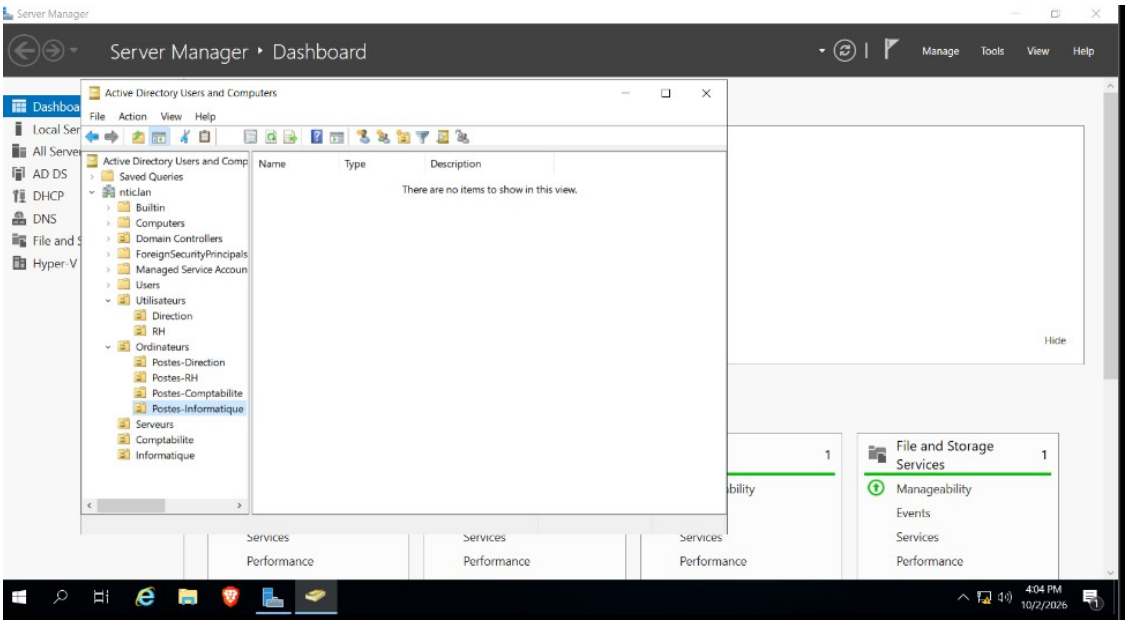
Résultat final : RODC-SERVER (192.168.10.11) apparaît dans Server Manager sous le rôle AD DS.



A.3 Administration des OU — volet interface graphique

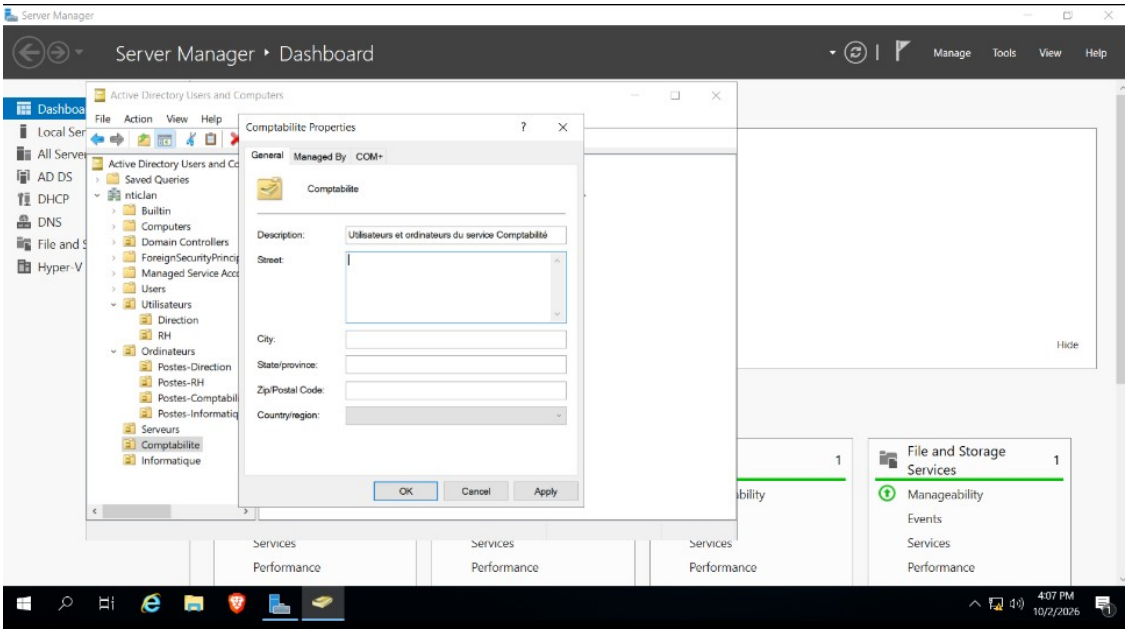
Travail demandé : création d'une arborescence d'unités d'organisation, d'un utilisateur et d'un ordinateur, réalisée ici via Active Directory Utilisateurs et ordinateurs (ADUC).

Structure obtenue (Utilisateurs > Direction/RH, Ordinateurs > Postes-*)

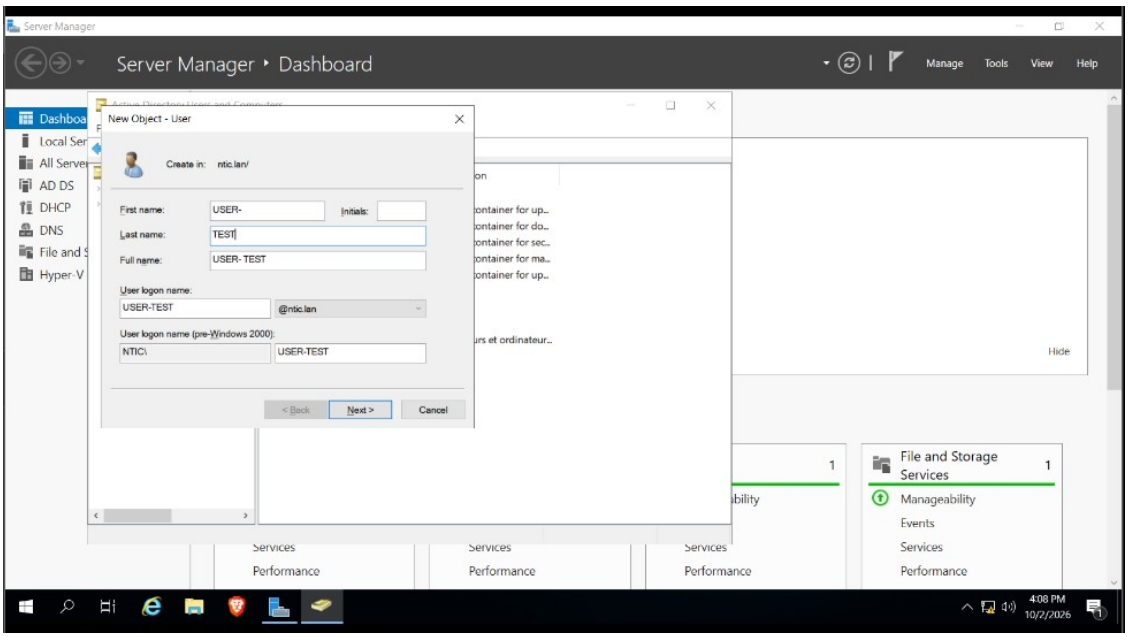


⚠️ Comptabilite et Informatique ont été créées à la racine du domaine plutôt que dans l'OU Utilisateurs.

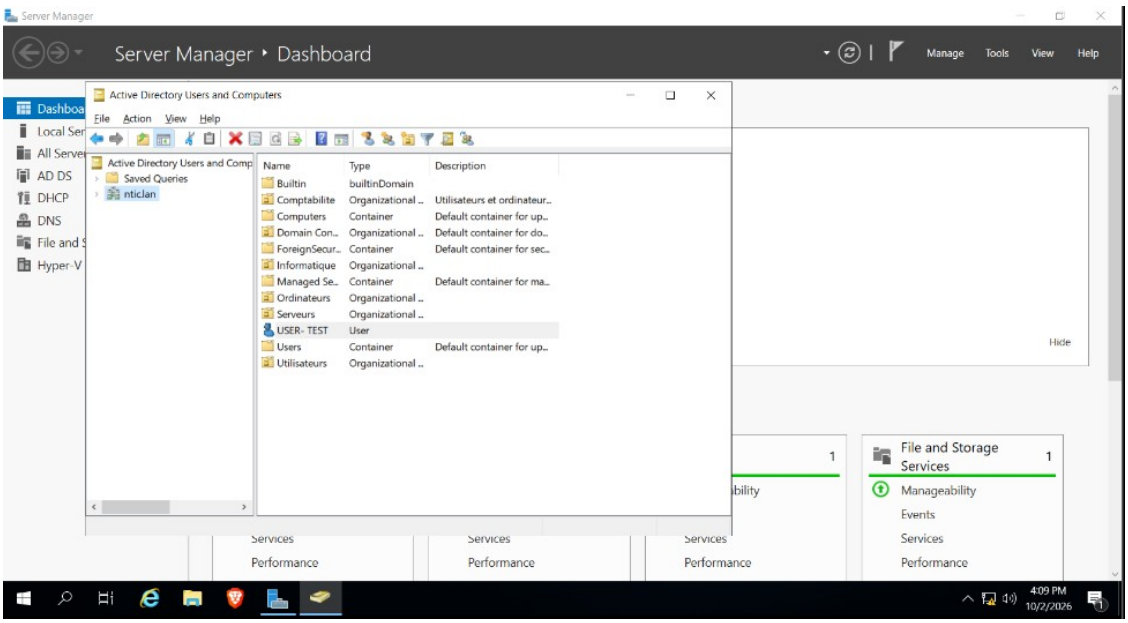
Description ajoutée à l'OU Comptabilite



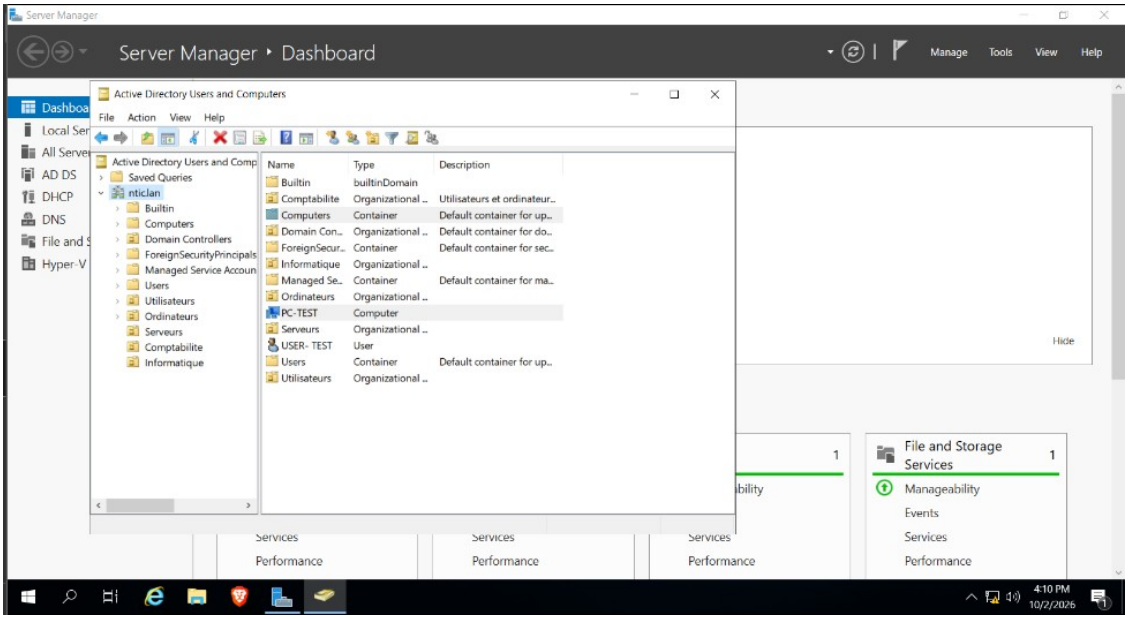
Création de l'utilisateur USER-TEST (assistant graphique)



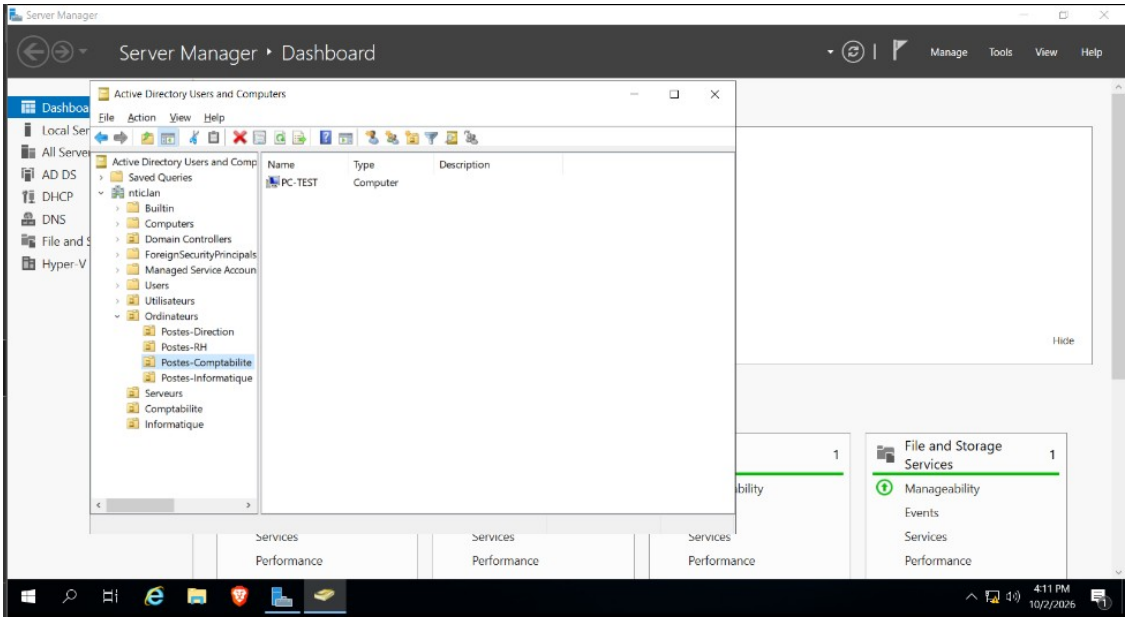
Vérification : le compte est bien créé (ici à la racine du domaine, avant déplacement).



Création de l'ordinateur PC-TEST (interface graphique)

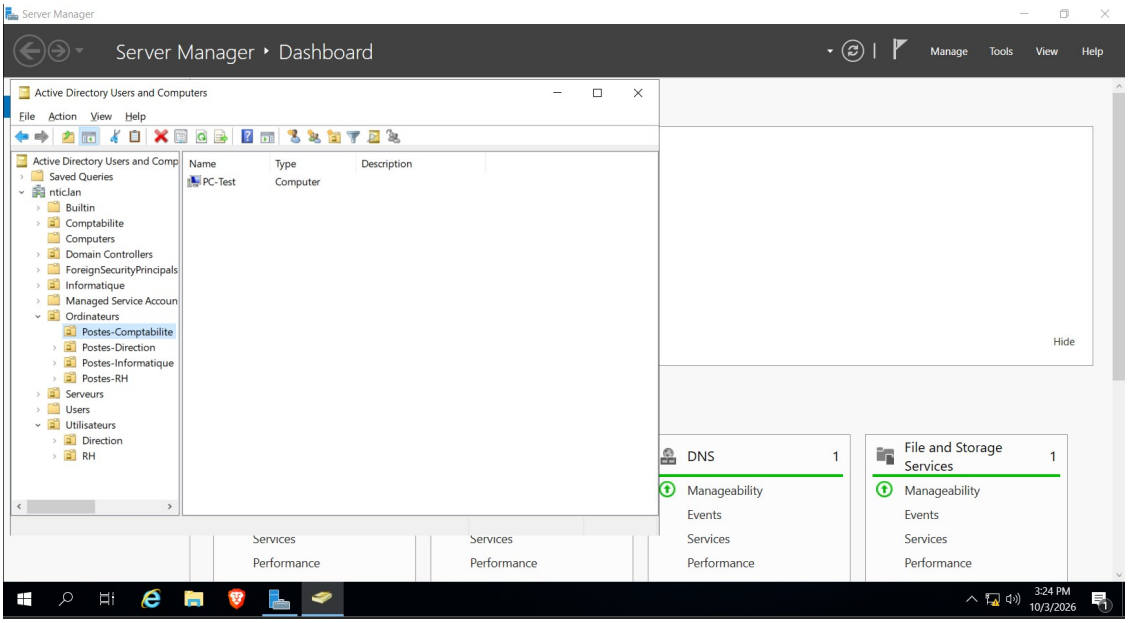


Déplacement initial de PC-TEST vers l'OU Postes-Comptabilite via l'interface graphique.



Arborescence finale (ADUC)

PC-TEST est bien positionné dans Postes-Comptabilite.

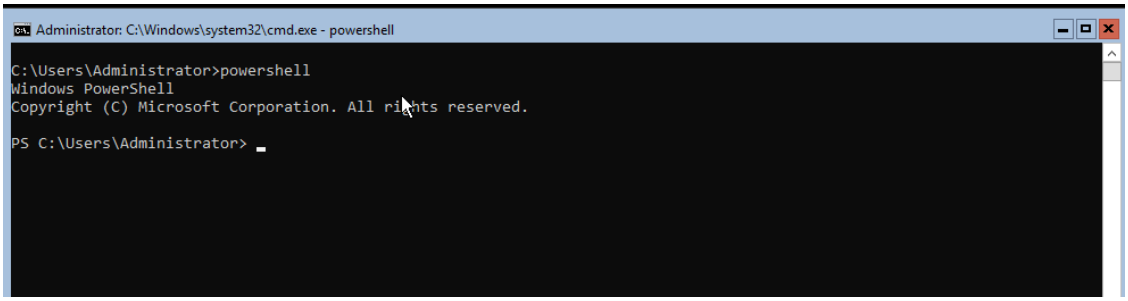


PARTIE B — LIGNE DE COMMANDE / POWERSHELL (SERVER CORE)

B.1 Mise en place d'un contrôleur de domaine avec PowerShell

1.1 Ouvrir une session PowerShell

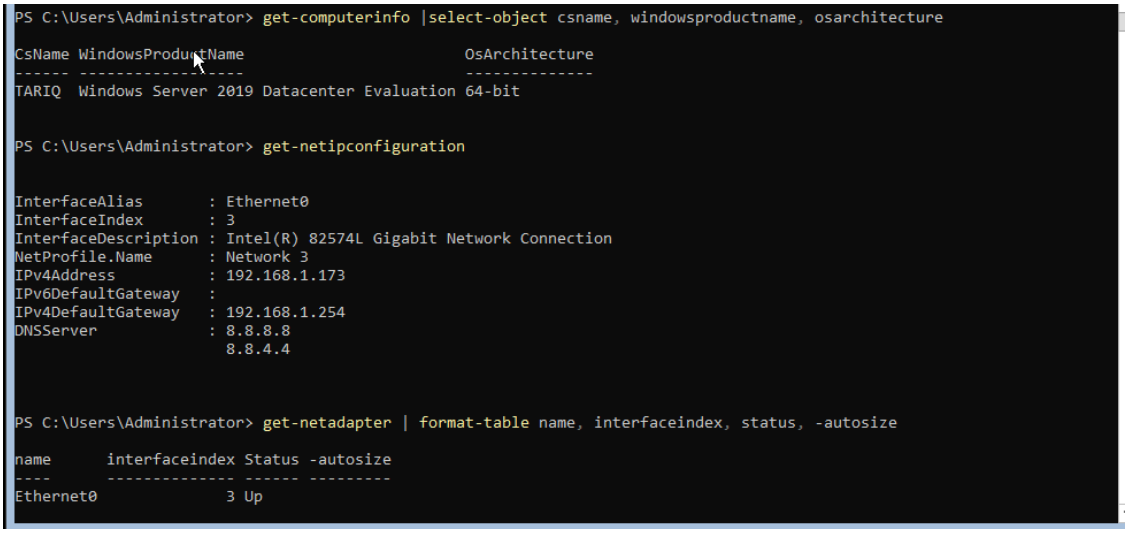
```
powershell
```



1.2 Identifier la configuration actuelle

```
hostname
Get-ComputerInfo | Select-Object CsName, WindowsProductName, OsArchitecture
Get-NetIPConfiguration
Get-NetAdapter | Format-Table Name, InterfaceIndex, Status -AutoSize
```





1.3 Renommer le serveur

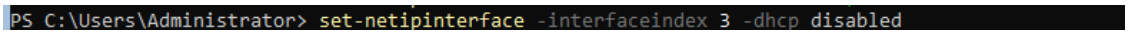
```
Rename-Computer -NewName "SRV-DC01" -Restart
```

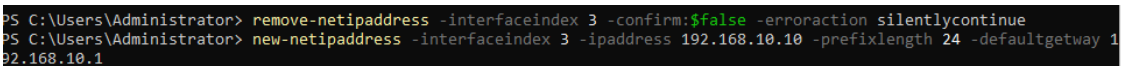


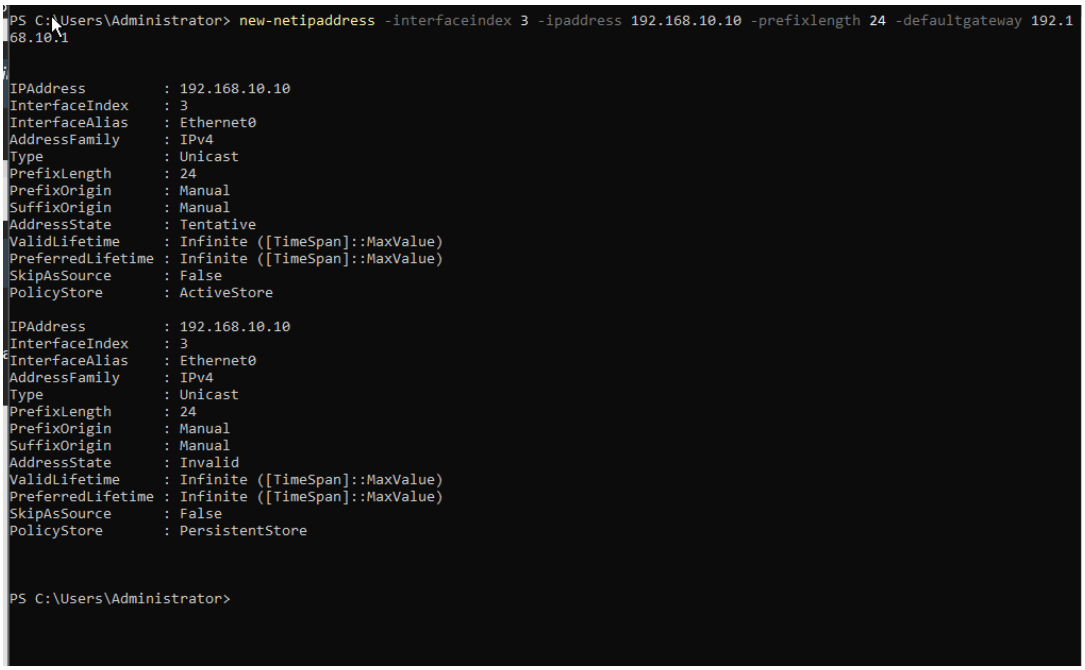
1.4 Configurer une adresse IP statique

```
Set-NetIPInterface -InterfaceIndex 4 -Dhcp Disabled
Remove-NetIPAddress -InterfaceIndex 4 -Confirm:$false -ErrorAction SilentlyContinue
New-NetIPAddress -InterfaceIndex 4 -IPAddress 192.168.10.10 -PrefixLength 24 -
DefaultGateway 192.168.10.1
```



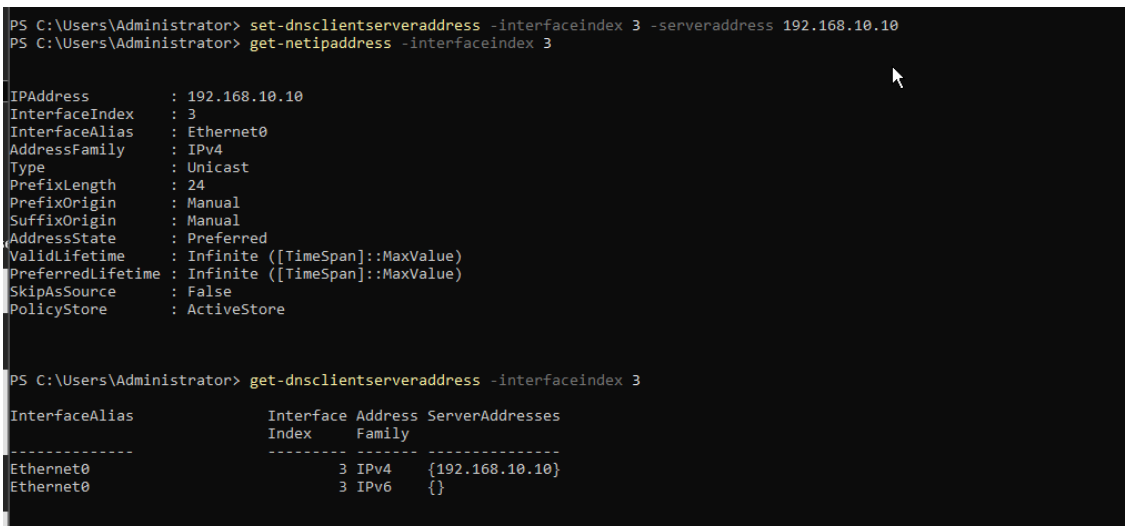






1.5 - 1.6 Configurer et vérifier le DNS

```
Set-DnsClientServerAddress -InterfaceIndex 4 -ServerAddresses 192.168.10.10
Get-NetIPAddress -InterfaceIndex 4
Test-Connection 192.168.10.1 -Count 2
```



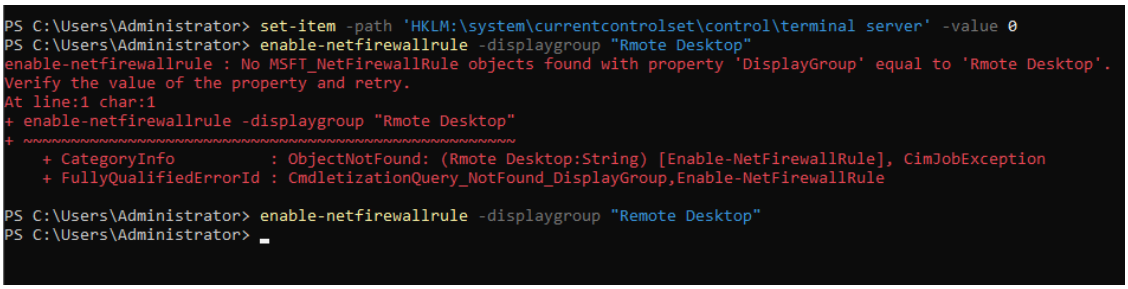
1.7 Fuseau horaire

```
Set-TimeZone -Id "Morocco Standard Time"
```

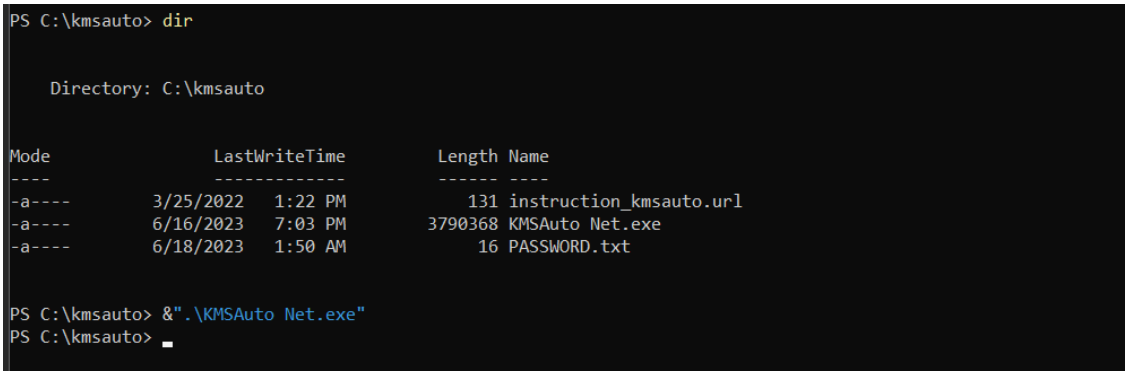


1.8 Activer le Bureau à distance

```
Set-ItemProperty -Path 'HKLM:\System\CurrentControlSet\Control\Terminal Server' -Name
"fDenyTSConnections" -Value 0
Enable-NetFirewallRule -DisplayGroup "Remote Desktop"
```



1.9 Activer le produit



1.10 Vérification finale

```
C:\Users\Administrator>cscript /nologo C:\Windows\System32\slmgr.vbs /xpr
Windows(R), ServerDatacenterEval edition:
    Timebased activation will expire 3/9/2027 8:32:03 PM


C:\Users\Administrator>powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Administrator> slmgr.vbs /dli
PS C:\Users\Administrator> hostname
SRV-DC01
PS C:\Users\Administrator> get-netipconfiguration


InterfaceAlias       : Ethernet0
InterfaceIndex       : 3
InterfaceDescription : Intel(R) 82574L Gigabit Network Connection
NetProfile.Name      : Unidentified network
IPv4Address          : 192.168.10.10
IPv6DefaultGateway   :
IPv4DefaultGateway   : 192.168.10.1
DNSServer            : fe80::1
                       192.168.10.10



PS C:\Users\Administrator> get-timezone


Id                         : Morocco Standard Time
DisplayName                : (UTC+01:00) Casablanca
StandardName               : Morocco Standard Time
DaylightName               : Morocco Daylight Time
BaseUtcOffset              : 00:00:00
SupportsDaylightSavingTime : True
```

2.2 Installer System Insights

```
Install-WindowsFeature -Name System-Insights -IncludeManagementTools
```

```
Administrator: C:\Windows\system32\cmd.exe - powershell
PS C:\Users\Administrator> Install-WindowsFeature -Name System-Insights -IncludeManagementTools

Success Restart Needed Exit Code      Feature Result
------- -------------- ---------      --------------
True    No             Success        {Remote Server Administration Tools, Featu...

PS C:\Users\Administrator> get-windowsfeature -name system-insights

Display Name                                            Name                    Install State
------------                                            ----                    -------------
[X] System Insights                                     System-Insights             Installed


PS C:\Users\Administrator> get-command -module systeminsights

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Cmdlet          Add-InsightsCapability                             1.0.0.0    systeminsights
Cmdlet          Disable-InsightsCapability                         1.0.0.0    systeminsights
Cmdlet          Disable-InsightsCapabilitySchedule                 1.0.0.0    systeminsights
Cmdlet          Enable-InsightsCapability                          1.0.0.0    systeminsights
Cmdlet          Enable-InsightsCapabilitySchedule                  1.0.0.0    systeminsights
Cmdlet          Get-InsightsCapability                             1.0.0.0    systeminsights
Cmdlet          Get-InsightsCapabilityAction                       1.0.0.0    systeminsights
Cmdlet          Get-InsightsCapabilityResult                       1.0.0.0    systeminsights
Cmdlet          Get-InsightsCapabilitySchedule                     1.0.0.0    systeminsights
Cmdlet          Invoke-InsightsCapability                          1.0.0.0    systeminsights
Cmdlet          Remove-InsightsCapability                          1.0.0.0    systeminsights
Cmdlet          Remove-InsightsCapabilityAction                    1.0.0.0    systeminsights
Cmdlet          Set-InsightsCapabilityAction                       1.0.0.0    systeminsights
Cmdlet          Set-InsightsCapabilitySchedule                     1.0.0.0    systeminsights
Cmdlet          Update-InsightsCapability                          1.0.0.0    systeminsights


PS C:\Users\Administrator>
```

3.1 - 3.2 Installer et vérifier le rôle AD DS

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
Get-WindowsFeature -Name AD-Domain-Services
```

```
PS C:\Users\Administrator> install-windowsfeature -name AD-Domain-Service -IncludeMangementTools
```

```
PS C:\Users\Administrator> Get-WindowsFeature -Name AD-Domain-Services

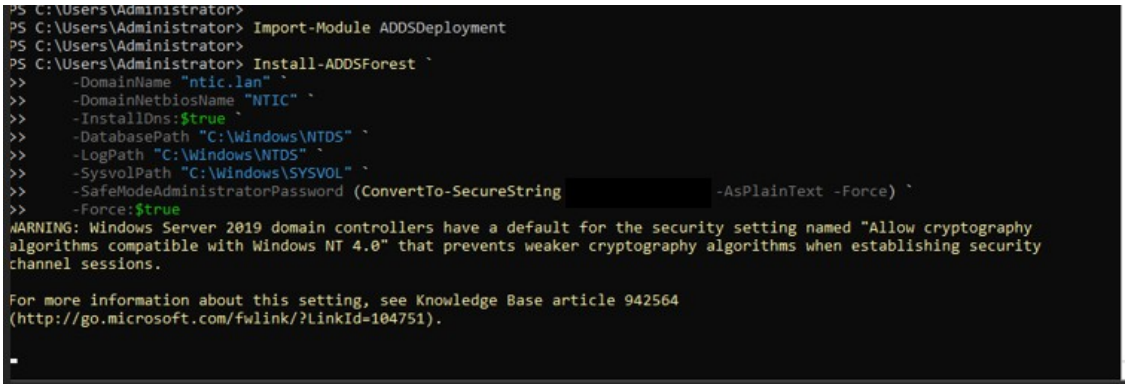
Display Name                                            Name                    Install State
------------                                            ----                    -------------
[X] Active Directory Domain Services                    AD-Domain-Services          Installed


PS C:\Users\Administrator>   Get-Command -Module ADDSDeployment

CommandType     Name                                               Version    Source
-----------     ----                                               -------    ------
Cmdlet          Add-ADDSReadOnlyDomainControllerAccount            1.0.0.0    ADDSDeployment
Cmdlet          Install-ADDSDomain                                 1.0.0.0    ADDSDeployment
Cmdlet          Install-ADDSDomainController                       1.0.0.0    ADDSDeployment
Cmdlet          Install-ADDSForest                                 1.0.0.0    ADDSDeployment
Cmdlet          Test-ADDSDomainControllerInstallation              1.0.0.0    ADDSDeployment
Cmdlet          Test-ADDSDomainControllerUninstallation            1.0.0.0    ADDSDeployment
Cmdlet          Test-ADDSDomainInstallation                        1.0.0.0    ADDSDeployment
Cmdlet          Test-ADDSForestInstallation                        1.0.0.0    ADDSDeployment
Cmdlet          Test-ADDSReadOnlyDomainControllerAccountCreation   1.0.0.0    ADDSDeployment
Cmdlet          Uninstall-ADDSDomainController                     1.0.0.0    ADDSDeployment
```

3.3 Promouvoir le serveur (nouvelle forêt ntic.lan)

```
Import-Module ADDSDeployment
Install-ADDSForest `
 -DomainName "ntic.lan" -DomainNetbiosName "NTIC" -InstallDns:$true `
 -SafeModeAdministratorPassword (ConvertTo-SecureString "<mot_de_passe>" -AsPlainText -
Force) -Force:$true
```



3.4 Vérifier la promotion et le DNS

```
Get-ADDomain
Get-ADForest
Get-Service NTDS, DNS, Netlogon | Select-Object Name, Status
Get-DnsServerZone
Resolve-DnsName srv-dc01.ntic.lan
```

```
PS C:\Users\Administrator> get-addomain


AllowedDNSSuffixes                : {}
ChildDomains                      : {}
ComputersContainer                : CN=Computers,DC=ntic,DC=lan
DeletedObjectsContainer           : CN=Deleted Objects,DC=ntic,DC=lan
DistinguishedName                 : DC=ntic,DC=lan
DNSRoot                           : ntic.lan
DomainControllersContainer        : OU=Domain Controllers,DC=ntic,DC=lan
DomainMode                        : Windows2016Domain
DomainSID                         : S-1-5-21-2916845633-1210579731-831391350
ForeignSecurityPrincipalsContainer : CN=ForeignSecurityPrincipals,DC=ntic,DC=lan
Forest                            : ntic.lan
InfrastructureMaster              : SRV-DC01.ntic.lan
LastLogonReplicationInterval      :
LinkedGroupPolicyObjects          : {CN={31B2F340-016D-11D2-945F-00C04FB984F9},CN=Policies,CN=System,DC=ntic,DC=lan}
LostAndFoundContainer             : CN=LostAndFound,DC=ntic,DC=lan
ManagedBy                         :
Name                              : ntic
NetBIOSName                       : NTIC
ObjectClass                       : domainDNS
ObjectGUID                        : 377db7ab-0343-4c22-a19d-a47cb06f4de5
ParentDomain                      :
PDCEmulator                       : SRV-DC01.ntic.lan
PublicKeyRequiredPasswordRolling  : True
QuotasContainer                   : CN=NTDS Quotas,DC=ntic,DC=lan
ReadOnlyReplicaDirectoryServers   : {}
ReplicaDirectoryServers           : {SRV-DC01.ntic.lan}
RIDMaster                         : SRV-DC01.ntic.lan
SubordinateReferences             : {CN=Configuration,DC=ntic,DC=lan}
SystemsContainer                  : CN=System,DC=ntic,DC=lan
UsersContainer                    : CN=Users,DC=ntic,DC=lan
```

```
PS C:\Users\Administrator> get-adforest


ApplicationPartitions : {}
CrossForestReferences : {}
DomainNamingMaster    : SRV-DC01.ntic.lan
Domains               : {ntic.lan}
ForestMode            : Windows2016Forest
GlobalCatalogs        : {SRV-DC01.ntic.lan}
Name                  : ntic.lan
PartitionsContainer   : CN=Partitions,CN=Configuration,DC=ntic,DC=lan
RootDomain            : ntic.lan
SchemaMaster          : SRV-DC01.ntic.lan
Sites                 : {Default-First-Site-Name}
SPNSuffixes           : {}
UPNSuffixes           : {}



PS C:\Users\Administrator> Get-Service NTDS, DNS, Netlogon | Select-Object Name, Status

Name      Status
----      ------
DNS       Running
Netlogon  Running
NTDS      Running


PS C:\Users\Administrator>   Get-DnsServerZone

ZoneName                          ZoneType       IsAutoCreated   IsDsIntegrated  IsReverseLookupZone  IsSigned
--------                          --------       -------------   --------------  -------------------  --------
_msdcs.ntic.lan                   Primary        False           True            False                False
0.in-addr.arpa                    Primary        True            False           True                 False
127.in-addr.arpa                  Primary        True            False           True                 False
255.in-addr.arpa                  Primary        True            False           True                 False
ntic.lan                          Primary        False           True            False                False
```

```
PS C:\Users\Administrator> Get-DnsServerResourceRecord -ZoneName "ntic.lan"

HostName                  RecordType Type       Timestamp           TimeToLive      RecordData
--------                  ---------- ----       ---------           ----------      ----------
@                         A          1          9/25/2026 10:00:0... 00:10:00        192.168.10.10
@                         NS         2          0                   01:00:00        srv-dc01.ntic.lan.
@                         SOA        6          0                   01:00:00        [19][srv-dc01.ntic.lan.][hostmast...
_msdcs                    NS         2          0                   01:00:00        srv-dc01.ntic.lan.
_gc._tcp.Default-First... SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][3268][SRV-DC01.ntic.lan.]
_kerberos._tcp.Default... SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][88][SRV-DC01.ntic.lan.]
_ldap._tcp.Default-Fir... SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][389][SRV-DC01.ntic.lan.]
_gc._tcp                  SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][3268][SRV-DC01.ntic.lan.]
_kerberos._tcp            SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][88][SRV-DC01.ntic.lan.]
_kpasswd._tcp             SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][464][SRV-DC01.ntic.lan.]
_ldap._tcp                SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][389][SRV-DC01.ntic.lan.]
_kerberos._udp            SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][88][SRV-DC01.ntic.lan.]
_kpasswd._udp             SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][464][SRV-DC01.ntic.lan.]
DomainDnsZones            A          1          9/25/2026 10:00:0... 00:10:00        192.168.10.10
_ldap._tcp.Default-Fir... SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][389][SRV-DC01.ntic.lan.]
_ldap._tcp.DomainDnsZones SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][389][SRV-DC01.ntic.lan.]
ForestDnsZones            A          1          9/25/2026 10:00:0... 00:10:00        192.168.10.10
_ldap._tcp.Default-Fir... SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][389][SRV-DC01.ntic.lan.]
_ldap._tcp.ForestDnsZones SRV        33         9/25/2026 10:00:0... 00:10:00        [0][100][389][SRV-DC01.ntic.lan.]
srv-dc01                  A          1          0                   01:00:00        192.168.10.10


PS C:\Users\Administrator> Resolve-DnsName srv-dc01.ntic.lan

Name                                           Type   TTL   Section    IPAddress
----                                           ----   ---   -------    ---------
SRV-DC01.ntic.lan                              AAAA   1200  Question   fe80::dc39:e52f:7e91:35cd
SRV-DC01.ntic.lan                              A      1200  Question   192.168.10.10


PS C:\Users\Administrator> nslookup ntic.lan
DNS request timed out.
    timeout was 2 seconds.
Server:  UnKnown
Address:  ::1

Name:    ntic.lan
Address:  192.168.10.10
```

3.5 Ordinateurs du domaine et création de etudiant1

```
Get-ADComputer -Filter * | Select-Object Name, DistinguishedName
New-ADUser -Name "Etudiant Un" -SamAccountName "etudiant1" -UserPrincipalName
"etudiant1@ntic.lan" ...
```

```
PS C:\Users\Administrator> Get-ADComputer -Filter * | Select-Object Name, DistinguishedName

Name            DistinguishedName
----            -----------------
SRV-DC01        CN=SRV-DC01,OU=Domain Controllers,DC=ntic,DC=lan
DESKTOP-BE0ODRB CN=DESKTOP-BE0ODRB,CN=Computers,DC=ntic,DC=lan


PS C:\Users\Administrator> Get-ADComputer -Identity "DESKTOP-BE0ODRB"


DistinguishedName : CN=DESKTOP-BE0ODRB,CN=Computers,DC=ntic,DC=lan
DNSHostName       : DESKTOP-BE0ODRB.ntic.lan
Enabled           : True
Name              : DESKTOP-BE0ODRB
ObjectClass       : computer
ObjectGUID        : 8e5d1495-72f7-46ce-ad8d-8868ccdc0c98
SamAccountName    : DESKTOP-BE0ODRB$
SID               : S-1-5-21-2916845633-1210579731-831391350-1103
UserPrincipalName :

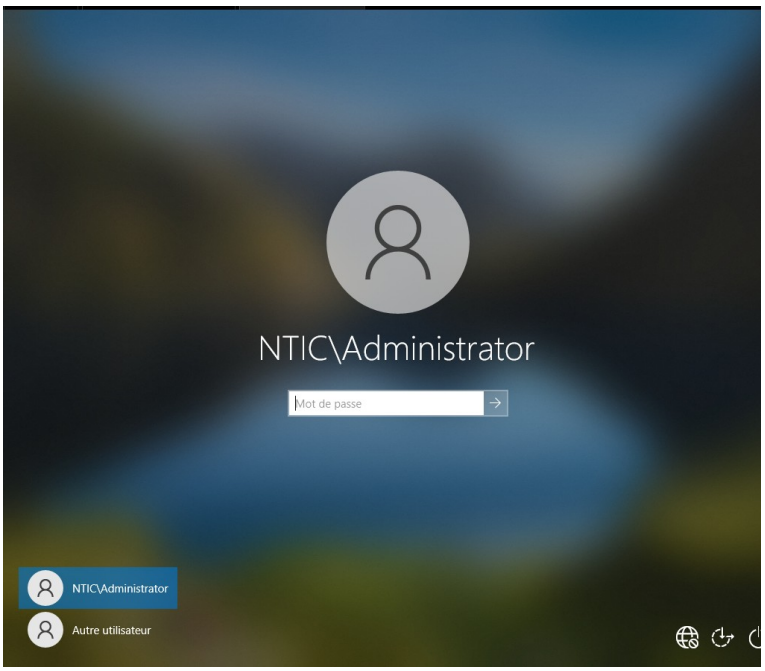


PS C:\Users\Administrator> New-ADUser -Name "Etudiant Un" -SamAccountName "etudiant1" `
>>     -UserPrincipalName "etudiant1@ntic.lan" `
>>     -AccountPassword (ConvertTo-SecureString                -AsPlainText -Force) `
>>     -Enabled $true
```

3.6 - 3.7 Canal sécurisé et écran de connexion

```
Testez le nouveau système multiplateforme PowerShell https://aka.ms/pscore6

PS C:\Users\Administrator>   whoami /fqdn
CN=Administrator,CN=Users,DC=ntic,DC=lan
PS C:\Users\Administrator> Test-ComputerSecureChannel -Verbose
COMMENTAIRES : Opération « Test-ComputerSecureChannel » en cours sur la cible « DESKTOP-BE0ODRB ».
True
COMMENTAIRES : Le canal sécurisé entre l'ordinateur local et le domaine ntic.lan fonctionne correctement.
PS C:\Users\Administrator>
```

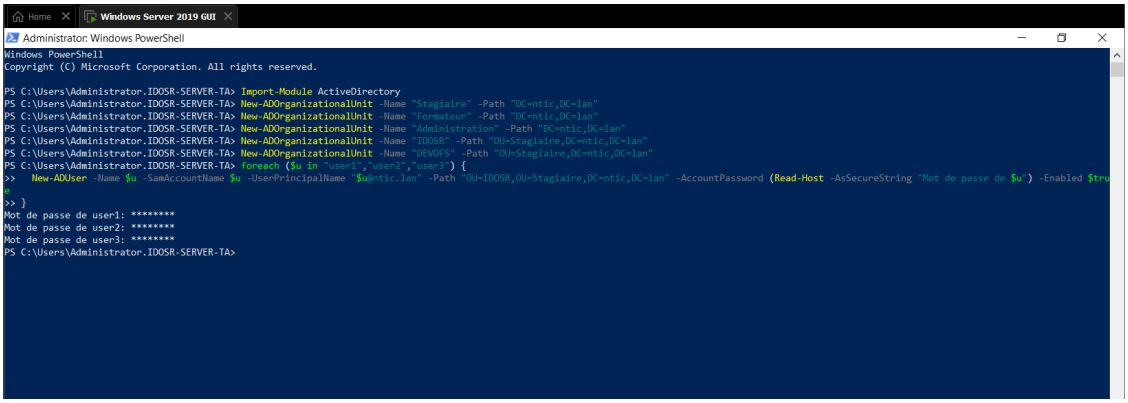


B.2 TP N°9 — Gérer les objets Active Directory en PowerShell

Question 1

Créer l'arborescence en lignes de commandes ; créer 3 utilisateurs user1, user2, user3 dans l'OU IDOSR.

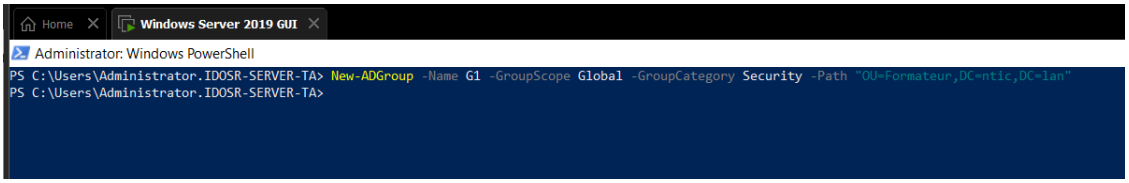
```
Import-Module ActiveDirectory
New-ADOrganizationalUnit -Name "Stagiaire" -Path "DC=ntic,DC=lan"
New-ADOrganizationalUnit -Name "Formateur" -Path "DC=ntic,DC=lan"
New-ADOrganizationalUnit -Name "Administration" -Path "DC=ntic,DC=lan"
New-ADOrganizationalUnit -Name "IDOSR" -Path "OU=Stagiaire,DC=ntic,DC=lan"
New-ADOrganizationalUnit -Name "DEVOFS" -Path "OU=Stagiaire,DC=ntic,DC=lan"
foreach ($u in "user1","user2","user3") {
 New-ADUser -Name $u -SamAccountName $u -UserPrincipalName "$u@ntic.lan" -Path
"OU=IDOSR,OU=Stagiaire,DC=ntic,DC=lan" -AccountPassword (Read-Host -AsSecureString "Mot
de passe de $u") -Enabled $true
}
```



Question 2

Créer le groupe G1 dans l'unité Formateur, étendue globale, type sécurité.

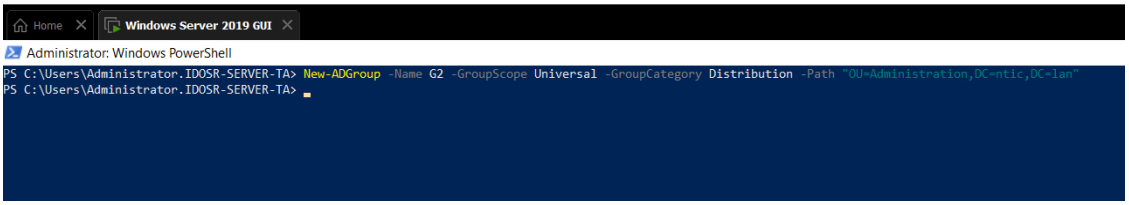
```
New-ADGroup -Name G1 -GroupScope Global -GroupCategory Security -Path
"OU=Formateur,DC=ntic,DC=lan"
```



Question 3

Créer le groupe G2 dans l'unité Administration, étendue universelle, type distribution.

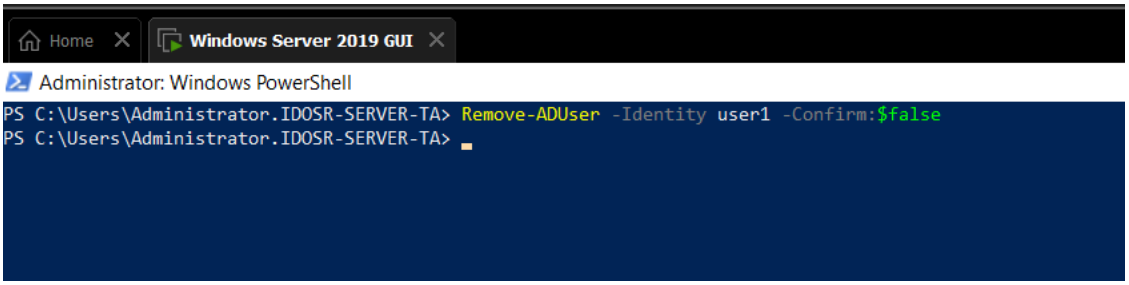
```
New-ADGroup -Name G2 -GroupScope Universal -GroupCategory Distribution -Path
"OU=Administration,DC=ntic,DC=lan"
```



Question 4

Supprimer l'utilisateur user1.

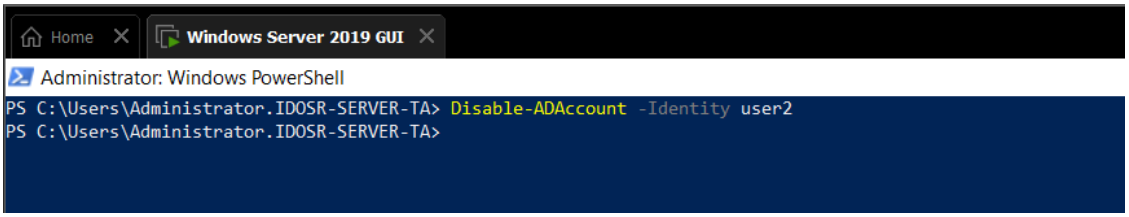
```
Remove-ADUser -Identity user1 -Confirm:$false
```



Question 5

Désactiver l'utilisateur user2.

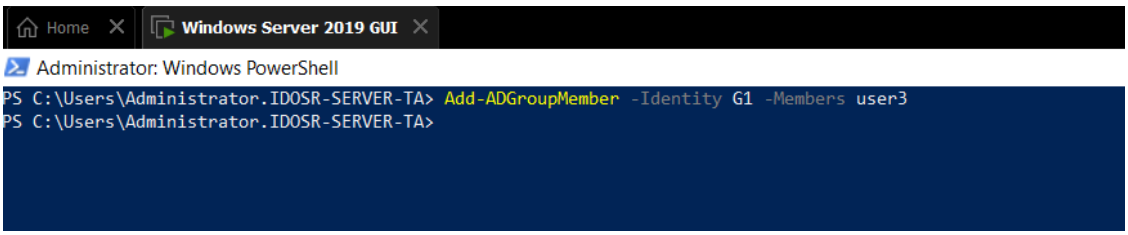
```
Disable-ADAccount -Identity user2
```



Question 6

Ajouter l'utilisateur user3 au groupe G1.

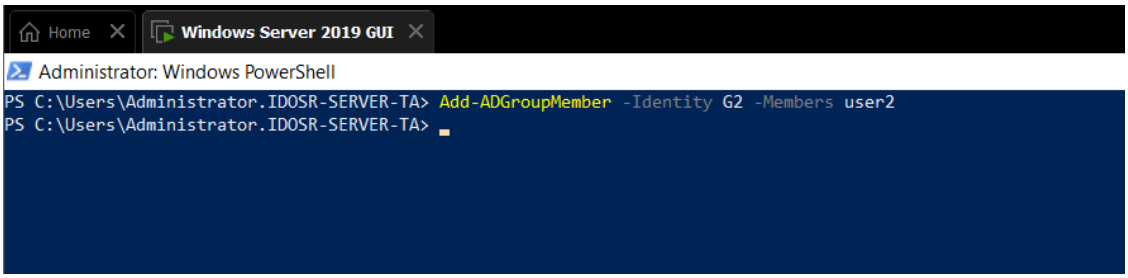
```
Add-ADGroupMember -Identity G1 -Members user3
```



Question 7

Affecter user2 au groupe G2.

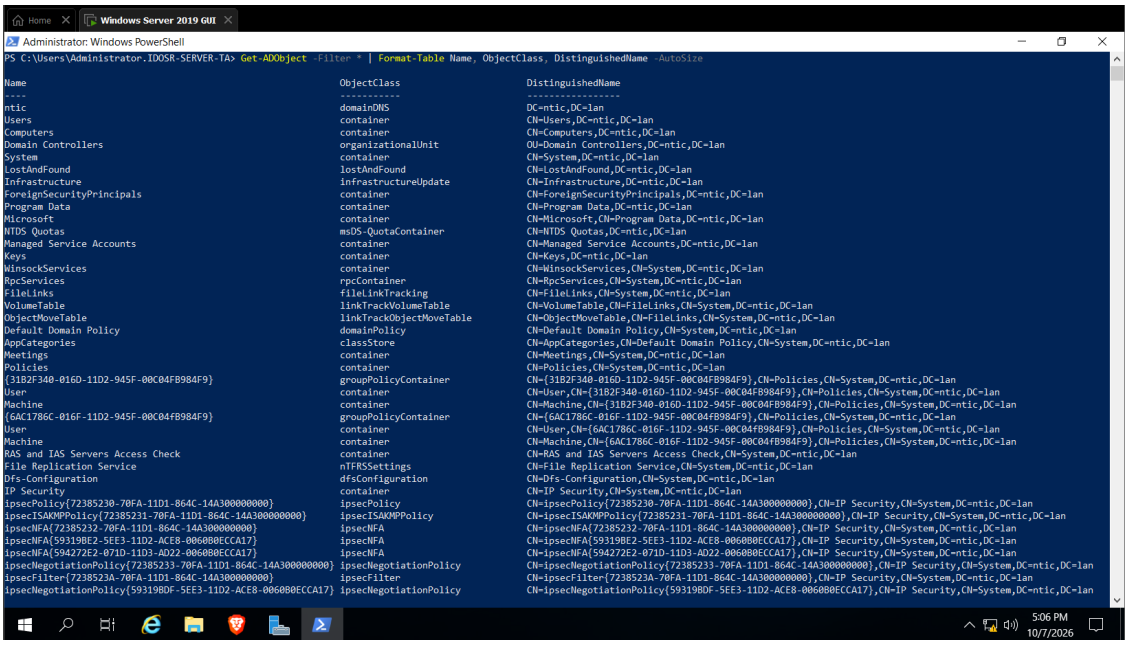
```
Add-ADGroupMember -Identity G2 -Members user2
```



Question 8

Afficher tous les objets AD.

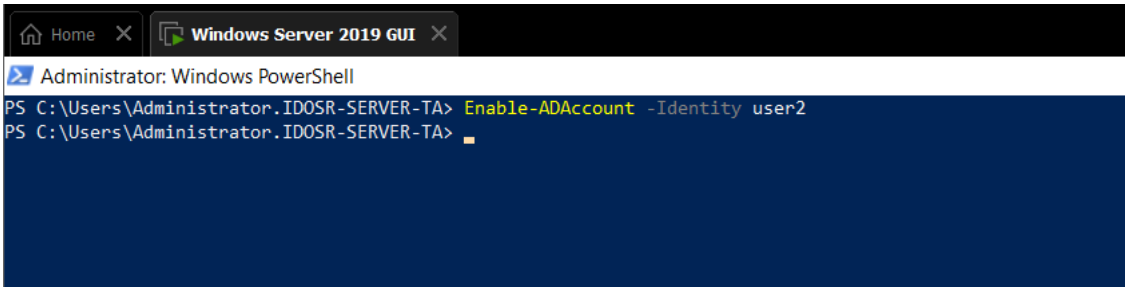
```
Get-ADObject -Filter * | Format-Table Name, ObjectClass, DistinguishedName -AutoSize
```



Question 9

Se connecter avec user2 : quel est le problème et comment le corriger ?

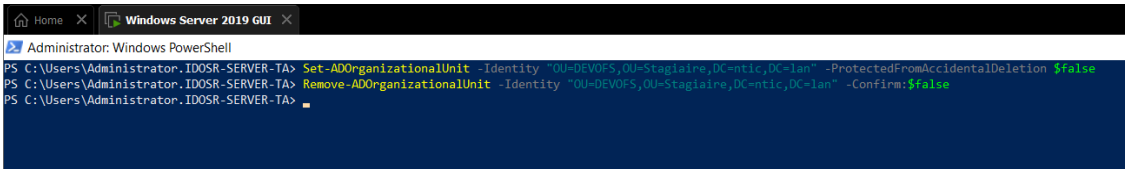
```
# Problème : le compte user2 est désactivé (question 5)
# Correction : Enable-ADAccount -Identity user2
```



Question 10

Supprimer l'unité d'organisation DEVOFS.

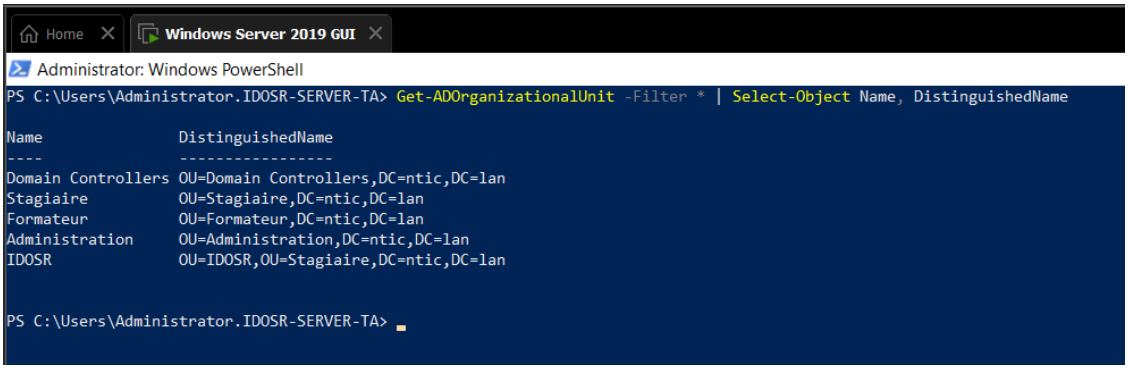
```
Set-ADOrganizationalUnit -Identity "OU=DEVOFS,OU=Stagiaire,DC=ntic,DC=lan" -
ProtectedFromAccidentalDeletion $false
Remove-ADOrganizationalUnit -Identity "OU=DEVOFS,OU=Stagiaire,DC=ntic,DC=lan" -Confirm:
$false
```



Question 11

Donner la nouvelle arborescence obtenue.

```
Get-ADOrganizationalUnit -Filter * | Select-Object Name, DistinguishedName
```



B.3 Administration des OU — volet PowerShell

1-4. Création des OU via PowerShell

Toutes les unités d'organisation (Utilisateurs, Ordinateurs, Serveurs, Comptabilite, Informatique, Direction, RH, Postes-) sont créées avec New-ADOrganizationalUnit, répondant à l'exigence « au moins une OU via PowerShell ».*



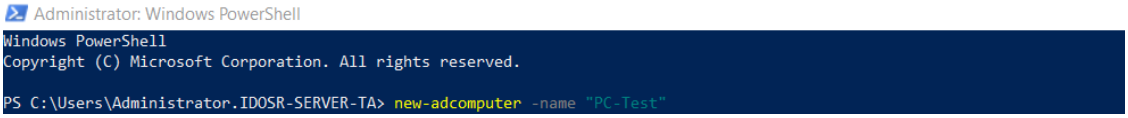
6. Déplacement de USER-TEST vers l'OU Comptabilite

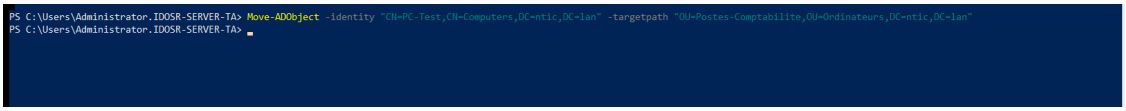
```
Move-ADObject -Identity "CN=User-Test,DC=ntic,DC=lan" -TargetPath
"OU=Comptabilite,DC=ntic,DC=lan"
```



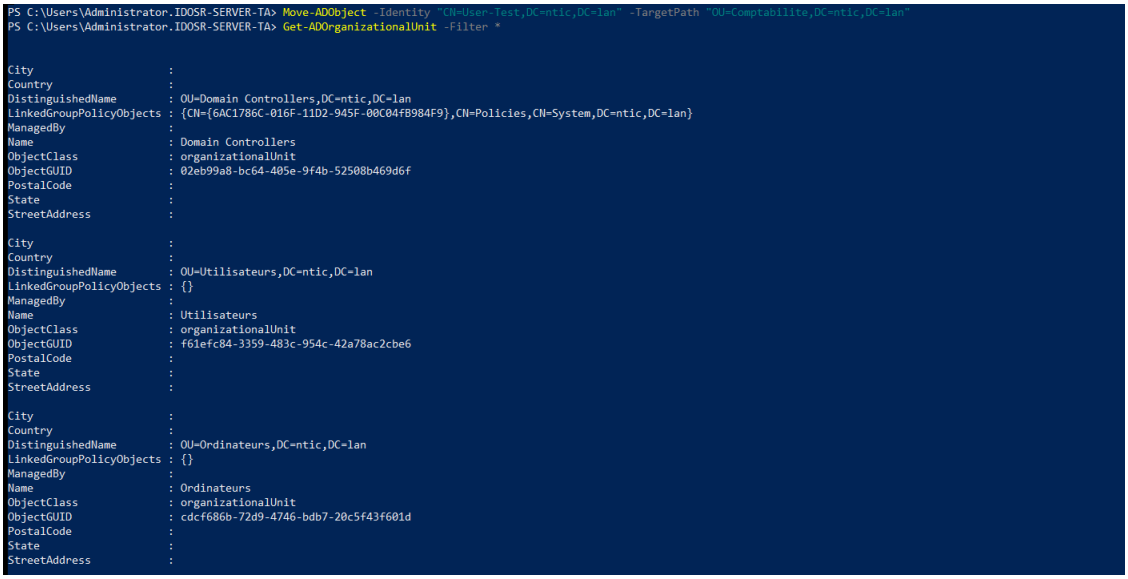
7. Recréation et déplacement de PC-TEST

Le compte PC-TEST est recréé via New-ADComputer puis déplacé vers Postes-Comptabilite avec Move-ADObject.





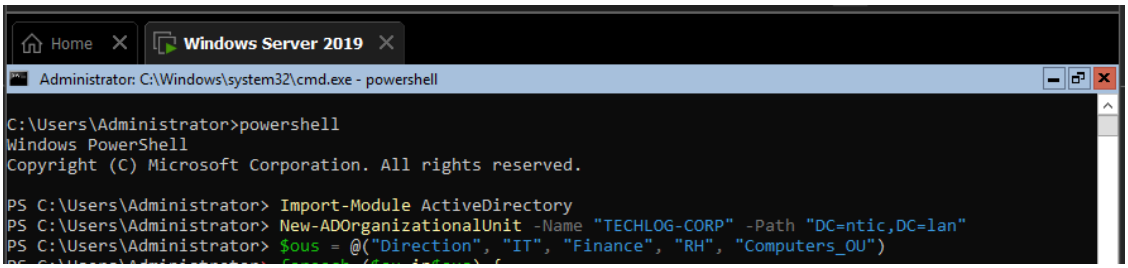
8. Vérification avec Get-ADOrganizationalUnit -Filter *



B.4 Création de groupes et utilisateurs — TECHLOG-CORP

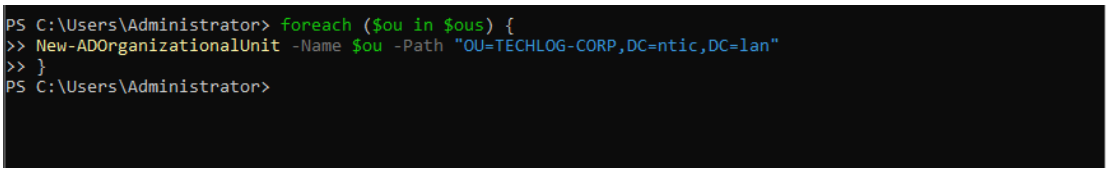
1. OU racine

```
New-ADOrganizationalUnit -Name "TECHLOG-CORP" -Path "DC=techlog,DC=local" `
 -ProtectedFromAccidentalDeletion $true
```



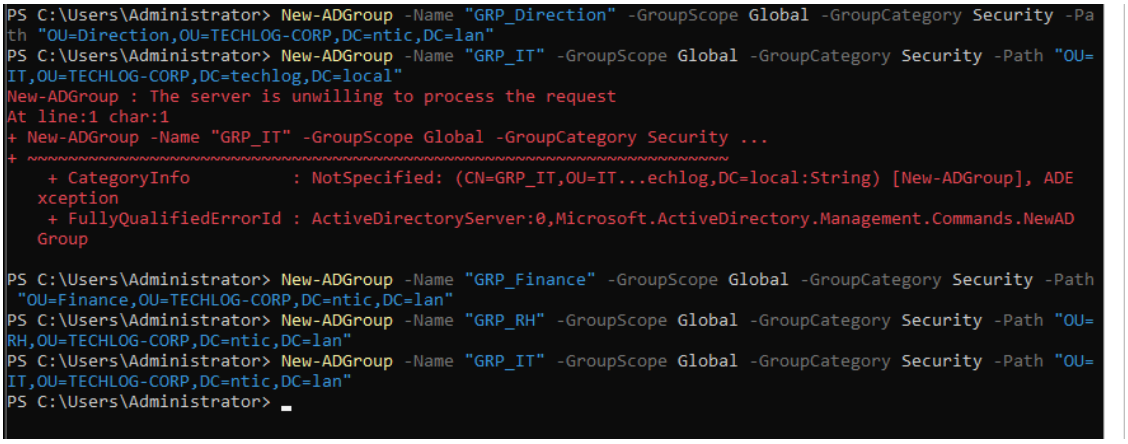
2. Sous-OU par département

```
$ous = @("Direction", "IT", "Finance", "RH", "Computers_OU")
foreach ($ou in $ous) {
  New-ADOrganizationalUnit -Name $ou -Path "OU=TECHLOG-CORP,DC=techlog,DC=local"
}
```



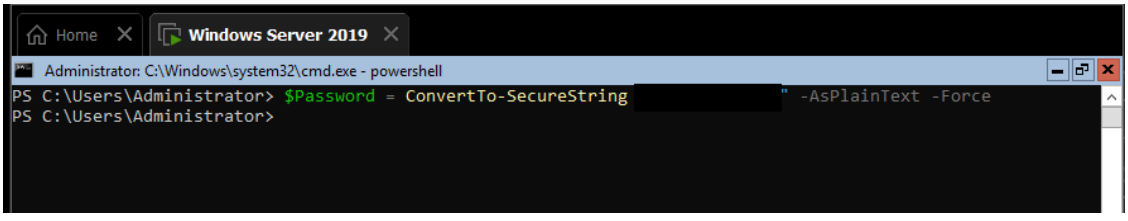
Groupes de sécurité par département

```
New-ADGroup -Name "GRP_Direction" -GroupScope Global -GroupCategory Security -Path
"OU=Direction,OU=TECHLOG-CORP,DC=techlog,DC=local"
New-ADGroup -Name "GRP_IT" ... -Path "OU=IT,..."
New-ADGroup -Name "GRP_Finance" ... -Path "OU=Finance,..."
New-ADGroup -Name "GRP_RH" ... -Path "OU=RH,..."
```

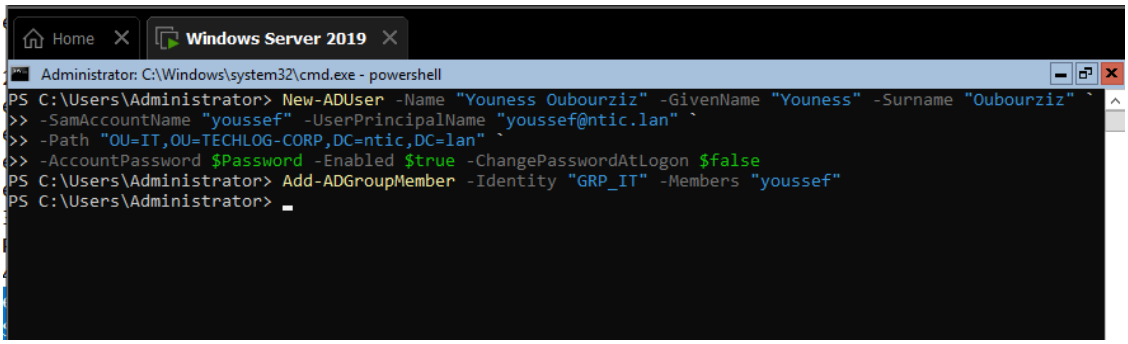


Création des utilisateurs et affectation aux groupes

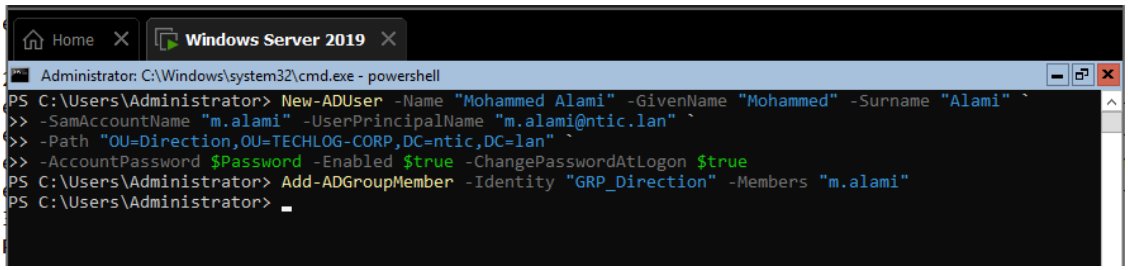
```
$Password = ConvertTo-SecureString "<mot_de_passe>" -AsPlainText -Force
```



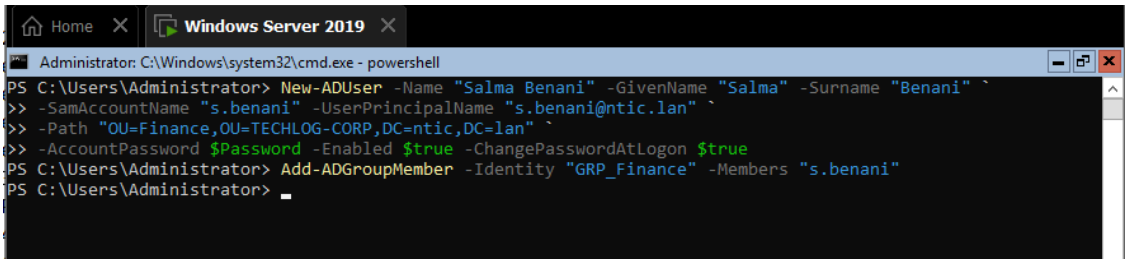
Youness Oubourziz (OU IT, GRP_IT)



Mohammed Alami (OU Direction, GRP_Direction)

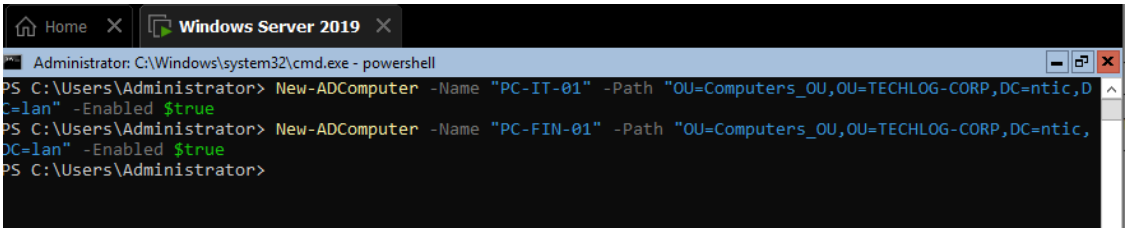


Salma Benani (OU Finance, GRP_Finance)



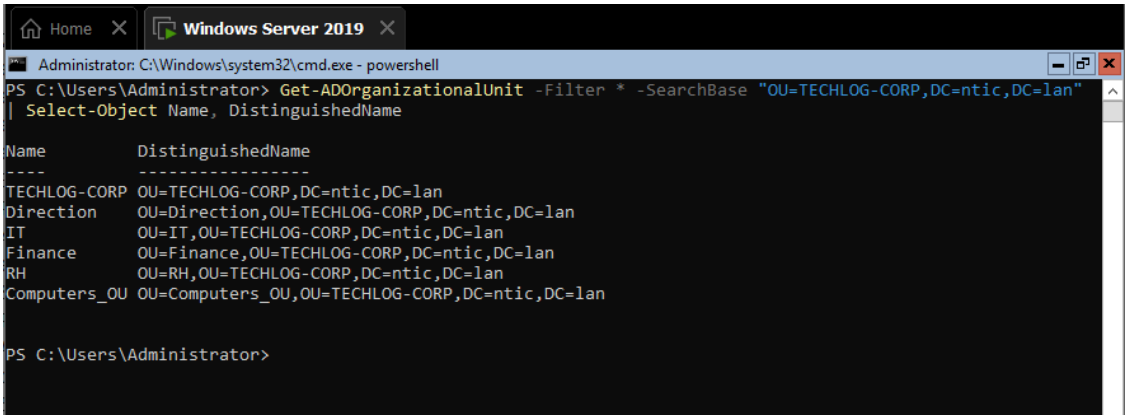
Pré-création des ordinateurs PC-IT-01 et PC-FIN-01

```
New-ADComputer -Name "PC-IT-01" -Path "OU=Computers_OU,OU=TECHLOG-
CORP,DC=techlog,DC=local" -Enabled $true
New-ADComputer -Name "PC-FIN-01" -Path "OU=Computers_OU,OU=TECHLOG-
CORP,DC=techlog,DC=local" -Enabled $true
```

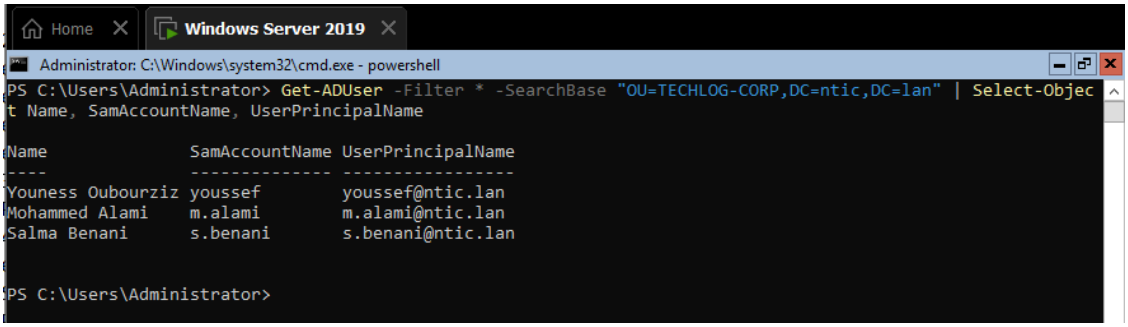


Vérifications globales

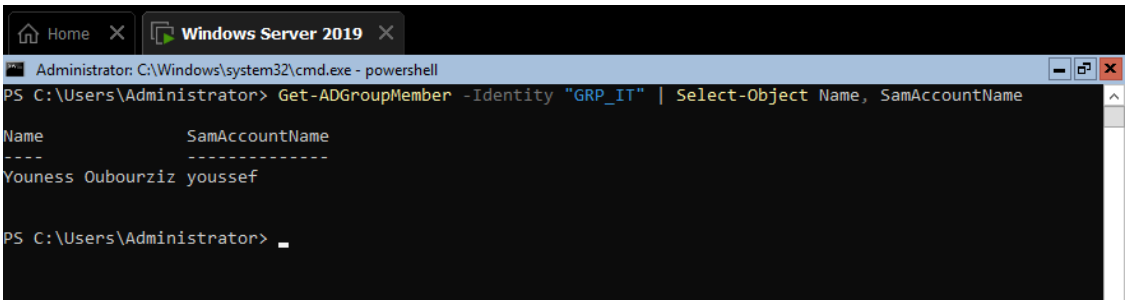
```
Get-ADOrganizationalUnit -Filter * -SearchBase "OU=TECHLOG-CORP,DC=techlog,DC=local" | Select-Object Name, DistinguishedName
```



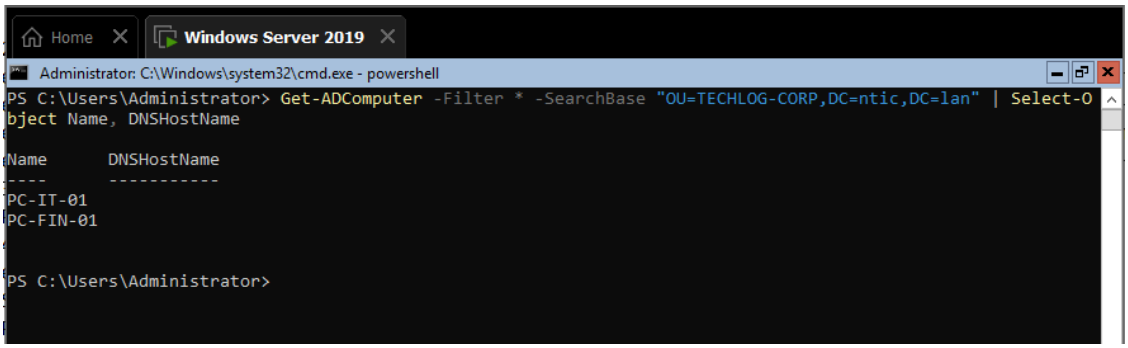

```
Get-ADUser -Filter * -SearchBase "OU=TECHLOG-CORP,DC=techlog,DC=local" | Select-Object Name, SamAccountName, UserPrincipalName
```




```
Get-ADGroupMember -Identity "GRP_IT" | Select-Object Name, SamAccountName
```




```
Get-ADComputer -Filter * -SearchBase "OU=TECHLOG-CORP,DC=techlog,DC=local" | Select-Object Name, DNSHostName
```



B.5 Réinstallation d'un RODC via PowerShell (Server Core)

Version en ligne de commande de la Partie A.2, pour le même scénario (serveur cloné) : rétrogradation, renommage, changement d'IP, puis réinstallation du RODC en rejoignant ntic.lan.

1. Supprimer AD DS du serveur cloné (rétrogradation)

```
Import-Module ADDSDeployment
Uninstall-ADDSDomainController `
 -DemoteOperationMasterRole -RemoveDnsDelegation `
 -LocalAdministratorPassword (ConvertTo-SecureString "<mot_de_passe>" -AsPlainText -
Force) -Force:$true
```

2. Renommer le serveur

```
Rename-Computer -NewName "RODC-SERVER" -Restart
```

3. Changer l'adresse IP

```
Remove-NetIPAddress -InterfaceAlias "Ethernet0" -Confirm:$false
New-NetIPAddress -InterfaceAlias "Ethernet0" -IPAddress 192.168.10.11 -PrefixLength 24 -
DefaultGateway 192.168.10.1
Set-DnsClientServerAddress -InterfaceAlias "Ethernet0" -ServerAddresses 192.168.10.10
```

4. Installer et promouvoir le RODC (rejoindre ntic.lan)

```
Install-WindowsFeature -Name AD-Domain-Services -IncludeManagementTools
Import-Module ADDSDeployment
Install-ADDSDomainController `
 -DomainName "ntic.lan" -ReadOnlyReplica -InstallDns:$true `
 -Credential (Get-Credential "NTIC\Administrator") `
 -DelegatedAdministratorAccountName "NTIC\etudiant1" `
 -SafeModeAdministratorPassword (ConvertTo-SecureString "<mot_de_passe>" -AsPlainText -
Force) -Force:$true
```

-Credential ouvre une invite pour saisir NTIC\Administrator + mot de passe du domaine.